

Intro: Files, BASH and Git

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Why?

and GitHub

Paleontological data in the 21st century

We have gone a long way...



Collectors only
-1960s



Pioneers
1960-1990/2000



Community of
database-based research

Being FAIR

A standard way to publish data and data-based research.

- Findable
- Accessible
- Interoperable
- Reproducible



www.go-fair.org

scientific **data**

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The FAIR Guiding Principles for scientific data management and stewardship

[Mark D. Wilkinson](#), [Michel Dumontier](#), ... [Barend Mons](#)  [+ Show authors](#)

[Scientific Data](#) **3**, Article number: 160018 (2016) | [Cite this article](#)

474k Accesses | **4409** Citations | **2001** Altmetric | [Metrics](#)

 An [Addendum](#) to this article was published on 19 March 2019

Reproducibility

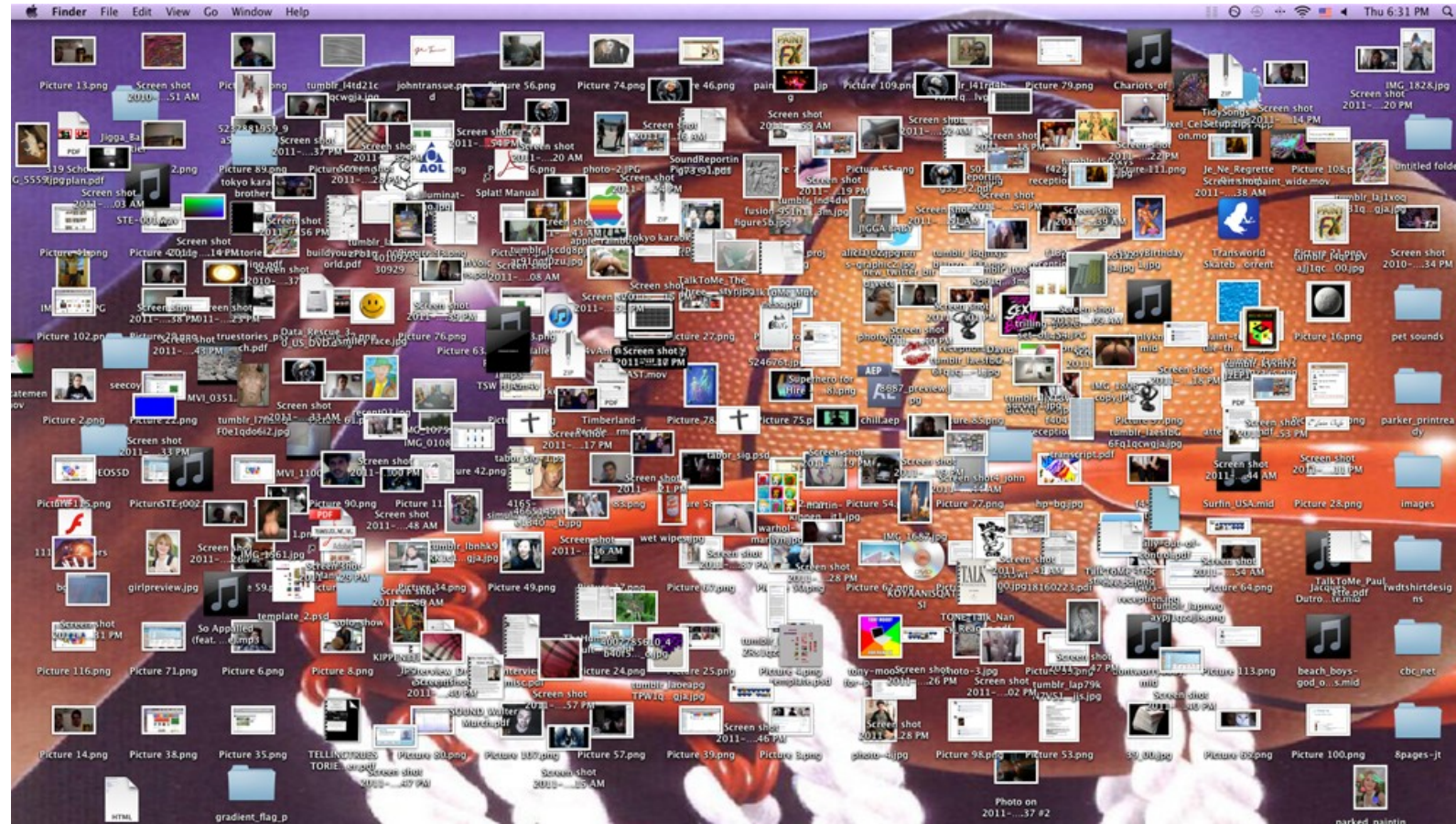
The foundation of the scientific experiment

- Can you reproduce the exact results that you acquired 5 years ago?
- If you cannot reproduce what you have done, how can other people?

Source: The Turing Way: <https://the-turing-way.netlify.app/>

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Avoid this at all costs...



Do not keep things on your desktop!

Overall file management

Suggestions

- Keep all your stuff together (separate partition!)
- Logical hierarchy
- Make it portable (Windows!)
- Regularly spend time on organizing and cleaning files
- Naming and grouping: self-explanatory – make it for somebody else (you!)
- Try to avoid spaces in paths
- Cloud backups!

Reproducibility is your main goal!

A Guide to
Reproducible Code
in Ecology and
Evolution



Suggestions

Keep all your projects separate!

Use the same project structure:

- Input Data (data)
- Computer code (code/scripts)
- Written documents (doc)
- Calculation output (export/output)

/ > media > adam > work > Dropbox > **WorkSpace**

Name	Size	Modified
2021-10-12_thermalSelect	5 items	27.06.22 08:01
2021-11-20_ordovician-bioge	1 item	20.11.21 11:42
2021-11-26_habitat	7 items	19.08.22 14:30
.git	11 items	15.07.22 22:51
data	4 items	07.07.22 11:34
doc	14 items	08.07.22 16:37
export	17 items	21.07.22 10:16
scripts	12 items	21.07.22 15:49
.gitignore	11 B	03.12.21 17:11
.projectile	0 B	29.11.21 17:24
2021-12-09_patterson	6 items	27.05.22 14:47
2021-12-10_BI	1 item	10.12.21 08:02
2022-01-14_datasyntesis	1 item	14.01.22 18:44
2022-03-01_bioDeepTime	13 items	04.08.22 15:17
2022-03-20_parameters	3 items	20.03.22 18:13

2021-10-12_thermalSelect (folder)

1,3 TiB free

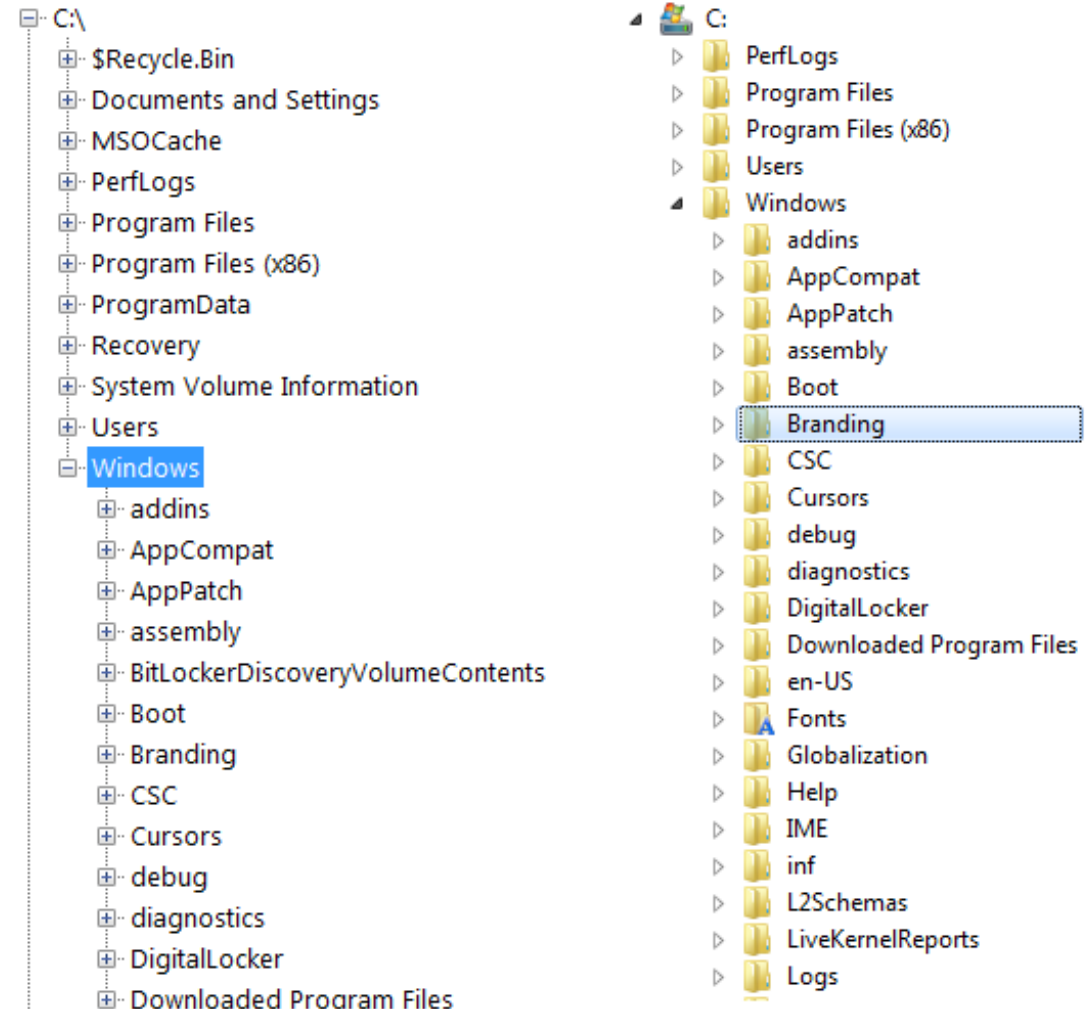
About files...

The Windows file system

- Files are data items on storage devices
- Paths use the characteristic backslash \ character to depict nestedness
- Directories are called “Folders”
- File format: filename.ext
- Total path to “Branding”:

C:\Windows\Branding

- Case insensitive!

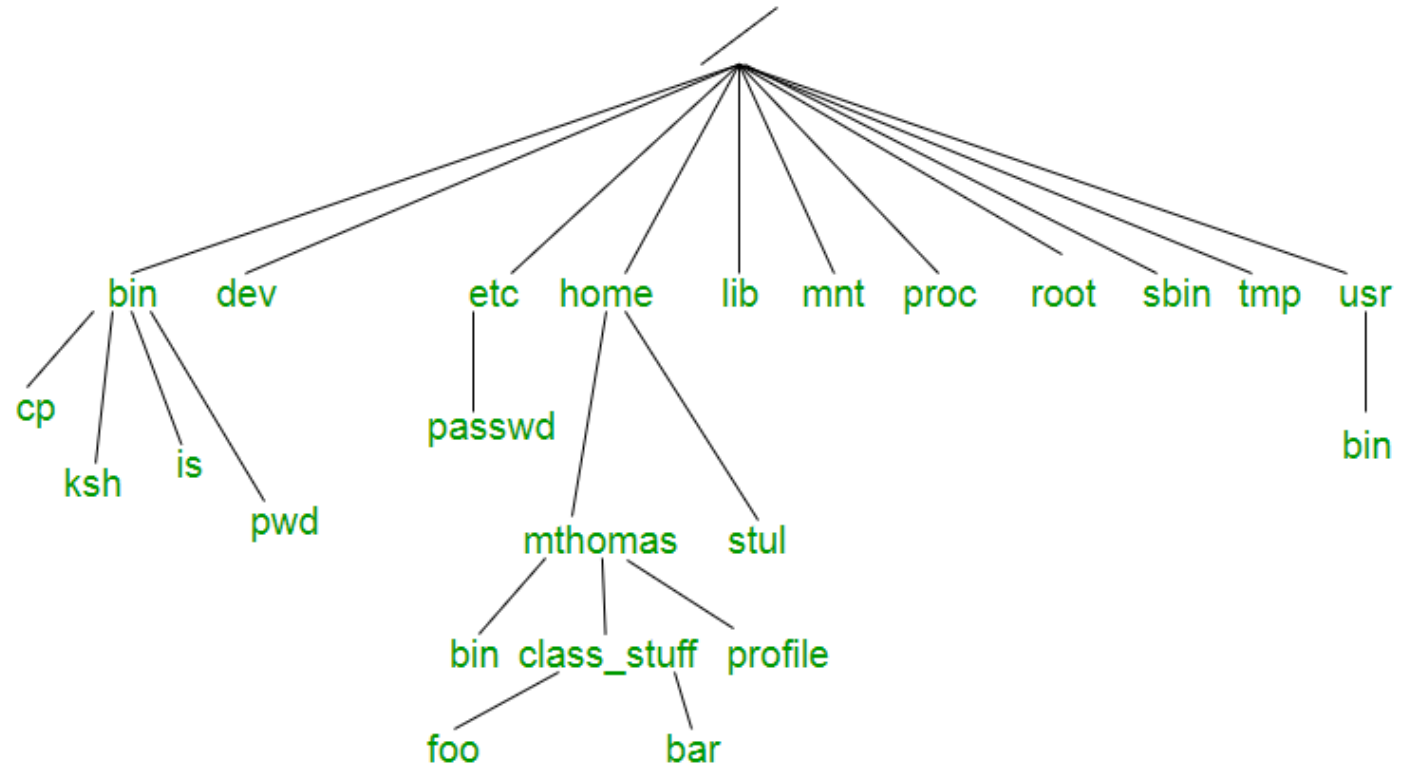


The UNIX file system

- Shared for UNIX and UNIX-like systems (GNU/Linux, macOS, Android)
- Concept: everything in the computer is represented by a file
- Nestedness coded with forward slash : /
- File format can be anything
- Complete path to “bar”

`/home/mthomas/class_stuff/bar`

- Case sensitive!



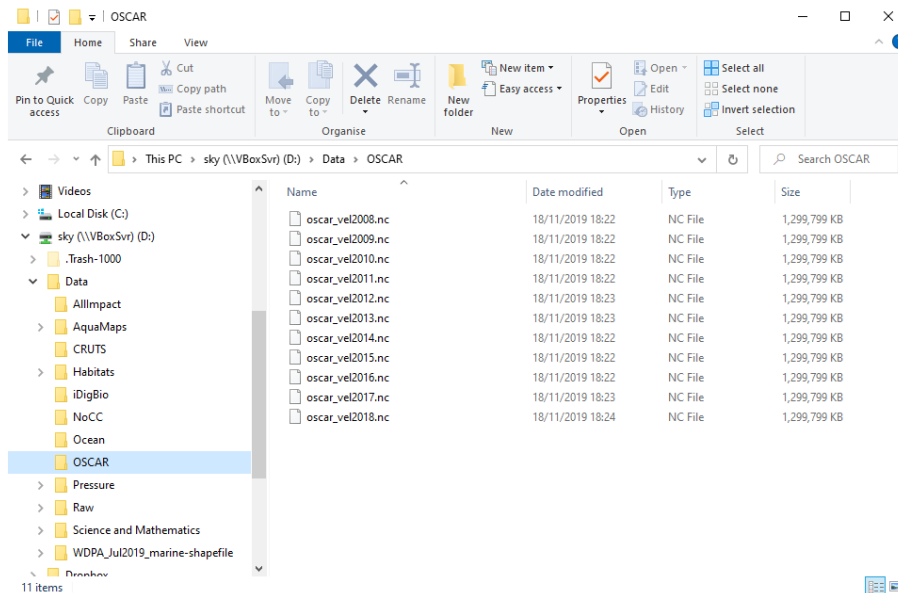
Two main options:

Graphical User Interface (GUI)

- Super simple + mouse
- Visually appealing
- “Novice-friendly”

Command Line Interpreter (CLI)

- Steeper learning curve
- Automation
- Keyboard-only “Expert-friendly”

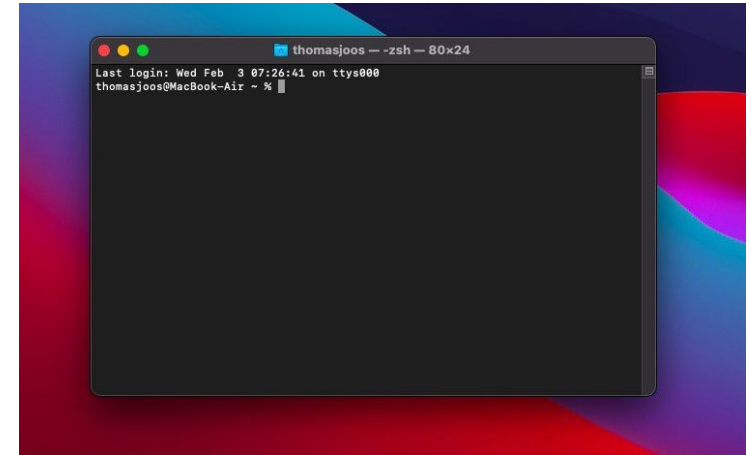


```
adam@posidonia: /mnt/sky/Data/OSCAR 154x19
adam@posidonia:~$ cd /mnt/sky/Data/OSCAR/
adam@posidonia:/mnt/sky/Data/OSCAR$ ls -la
total 14297852
drwxrwxr-x  2 adam adam    4096 Sep  4  2020 .
drwxrwxr-x 14 adam adam    4096 Okt 23  2021 ..
-rwxrwxrwx  1 adam adam 1330993460 Nov 18  2019 oscar_vel2008.nc
-rwxrwxrwx  1 adam adam 1330993460 Nov 18  2019 oscar_vel2009.nc
-rwxrwxrwx  1 adam adam 1330993460 Nov 18  2019 oscar_vel2010.nc
-rwxrwxrwx  1 adam adam 1330993460 Nov 18  2019 oscar_vel2011.nc
-rwxrwxrwx  1 adam adam 1330993512 Nov 18  2019 oscar_vel2012.nc
-rwxrwxrwx  1 adam adam 1330993512 Nov 18  2019 oscar_vel2013.nc
-rwxrwxrwx  1 adam adam 1330993512 Nov 18  2019 oscar_vel2014.nc
-rwxrwxrwx  1 adam adam 1330993512 Nov 18  2019 oscar_vel2015.nc
-rwxrwxrwx  1 adam adam 1330993512 Nov 18  2019 oscar_vel2016.nc
-rwxrwxrwx  1 adam adam 1330993512 Nov 18  2019 oscar_vel2017.nc
-rwxrwxrwx  1 adam adam 1330993512 Nov 18  2019 oscar_vel2018.nc
adam@posidonia:/mnt/sky/Data/OSCAR$
```

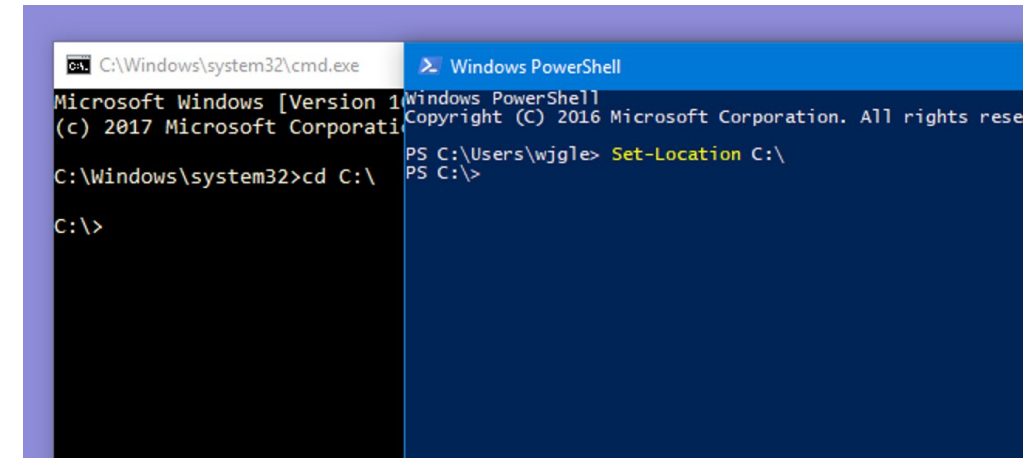

Terminal emulators

- Every OS has one
- Graphical applications that run a program called the “shell”: an interpreter program that translates instructions
- Console applications can be run with the shell
 - Automation
 - Program building
 - Scientific calculations
- Shells are programmable

Mac (zsh or bash)



Windows (cmd and powershell)



The BASH shell

- Ubiquitous
- Most frequently used on servers and clusters
- UNIX-native: most programming systems use UNIX-like paths – even on Windows!
- Mac: have it, z shell (zsh) is almost the same
- Windows: a simplified version is available with git (git bash)



<https://git-scm.com>

Installing git for Windows

and GitHub

BASH essentials

Most important functions and browsing directories

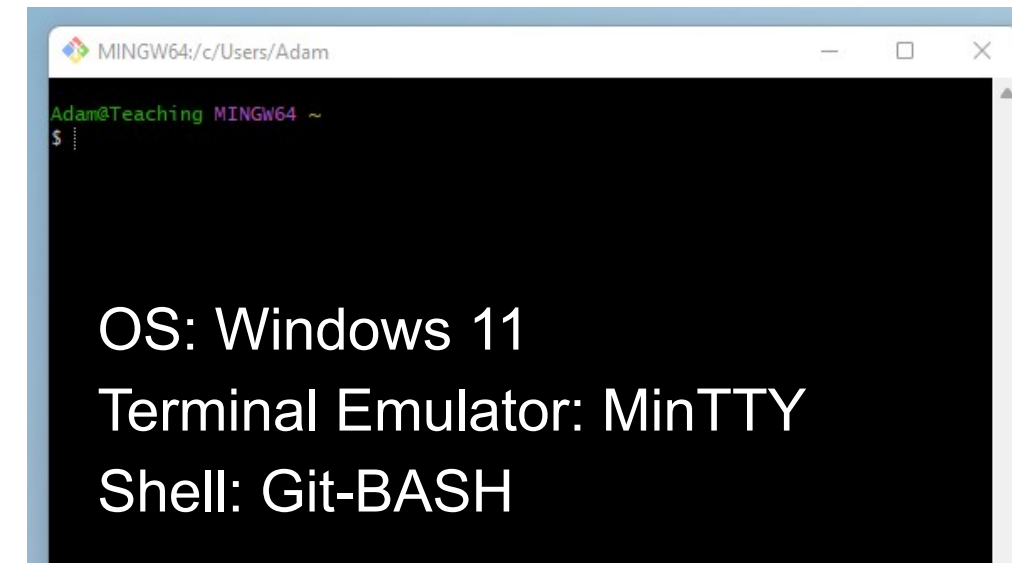
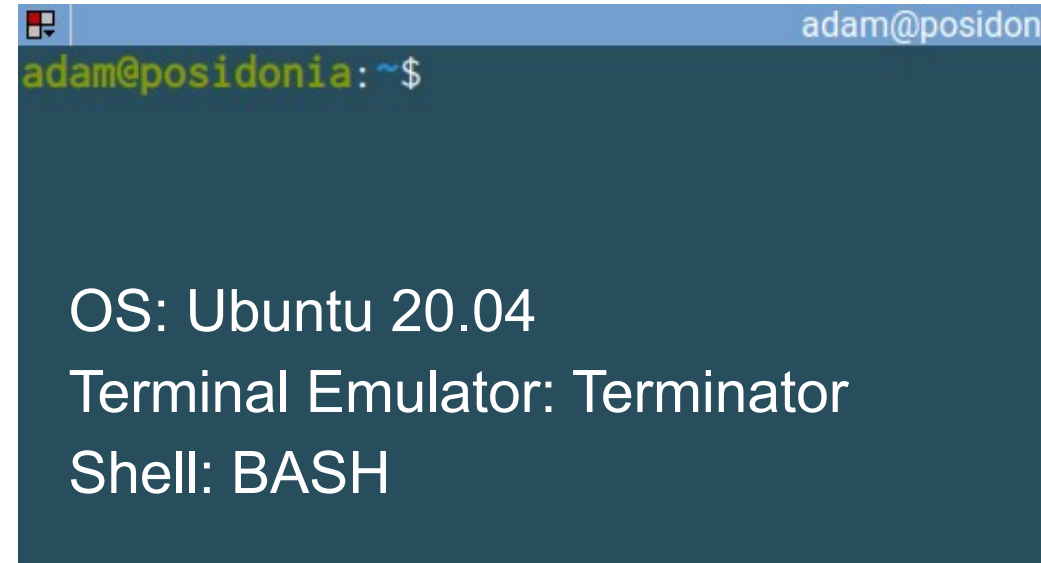
The prompt

- User input expected (typing)
- Looks different on all, but there are conventions:

`user@host`

`~`: is shorthand for user home

`$`: means normal user mode



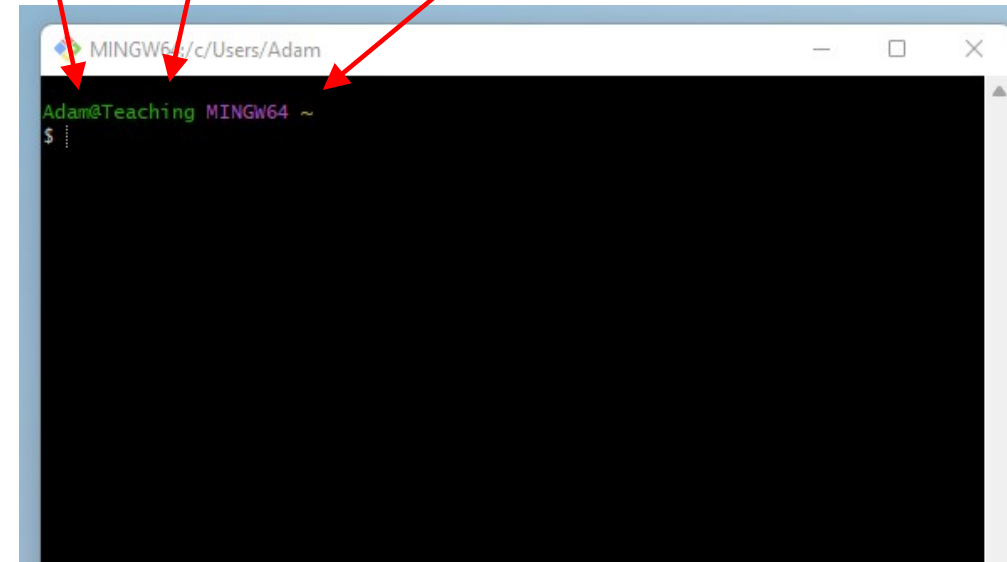
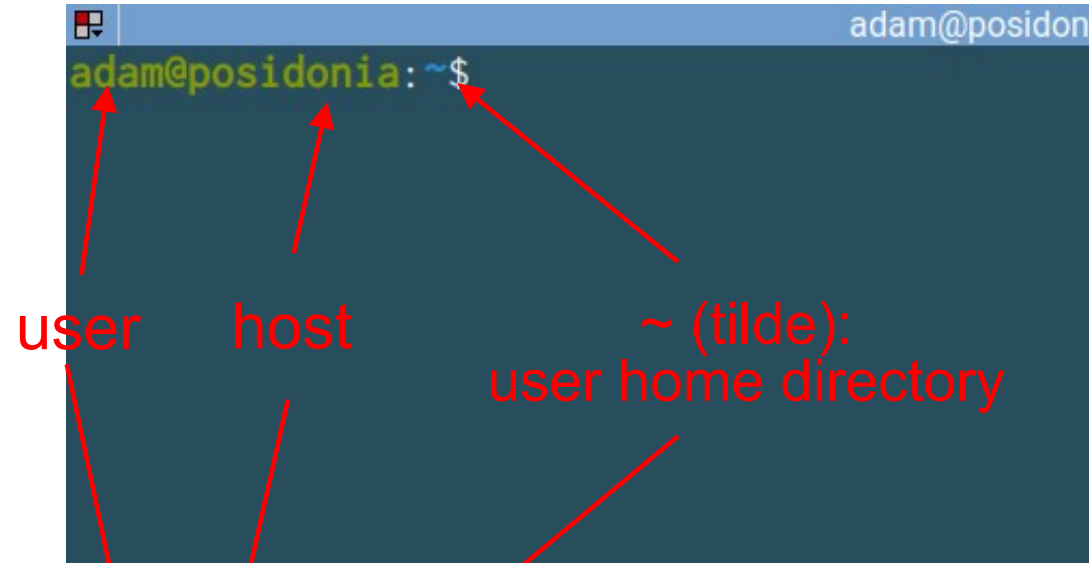
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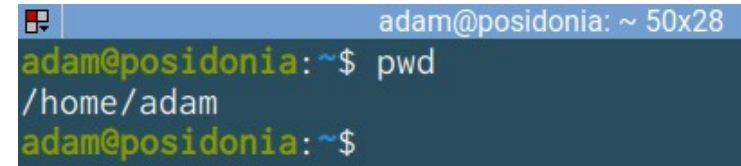
`~`: is shorthand for user home

`$`: means normal user mode



pwd

Return path to current directory



```
adam@posidonia: ~ 50x28
adam@posidonia:~$ pwd
/home/adam
adam@posidonia:~$
```

A terminal window with a blue title bar containing the text "adam@posidonia: ~ 50x28". The terminal has a dark blue background. The prompt "adam@posidonia:~\$" is shown in green. The command "pwd" is entered in white. The output "/home/adam" is shown in white. The prompt "adam@posidonia:~\$" is shown again in green.

`mkdir _<name>`

space

Create a directory

- No output to the console: no error occurred (directory was created)

```
adam@posidonia: ~ 50x28
adam@posidonia:~$ mkdir my_dir
adam@posidonia:~$
```

ls

List directory contents

- Returns a list of entries (both normal files and directories) – can be colored
- Note the quotes around entries with spaces in them!



```
adam@posidonia: ~ 50x28
adam@posidonia:~$ ls
01-network-manager-all.yaml  Programs
1_linktags.sh                Public
Desktop                      random.conf
Documents                    report
Downloads                    snap
Edraw                        some.df
exercises                    Templates
gems                        temp.mbsync
Mail                        Videos
Music                      virtual
my_dir                     'VirtualBox VMs'
'NVIDIA Nsight Systems'    zen.json
Pictures                  Zotero
adam@posidonia:~$
```

ls -l

List directory contents (with option l)

- Long output, includes attributes

d:directory

permissions

owner

size (bytes)

modification

name

```
adam@posidonia: ~ 73x30
adam@posidonia:~$ ls -l
total 112
-rw-r--r-- 1 adam adam 104 Sep  2  2020 01-network-manager-all.yaml
-rw-r--r-- 1 adam adam 224 Jul  2 13:34 1_linktags.sh
drwxr-xr-x 3 adam adam 4096 Jul 29 19:58 Desktop
drwxrwxr-x 2 adam adam 4096 Dez 16  2021 Documents
drwxr-xr-x 5 adam adam 16384 Aug 19 12:00 Downloads
drwxrwxr-x 3 adam adam 4096 Sep  5  2020 Edraw
drwxrwxr-x 3 adam adam 4096 Okt 14  2021 exercises
drwxrwxr-x 10 adam adam 4096 Aug 15 14:40 gems
drwxrwxr-x 7 adam adam 4096 Jan 24  2022 Mail
drwxr-xr-x 2 adam adam 4096 Sep  1  2020 Music
drwxrwxr-x 2 adam adam 4096 Aug 19 16:09 my_dir
drwxrwxr-x 2 adam adam 4096 Apr 21 15:34 'NVIDIA Nsight Systems'
drwxr-xr-x 2 adam adam 4096 Sep  1  2020 Pictures
drwxrwxr-x 4 adam adam 4096 Apr 28  2021 Programs
drwxr-xr-x 2 adam adam 4096 Sep  1  2020 Public
-rw-rw-r-- 1 adam adam  9 Aug 17 17:39 random.conf
drwxrwxr-x 3 adam adam 4096 Jun  2 12:36 report
drwx----- 5 adam adam 4096 Jun  4  2021 snap
drwxr-xr-x 2 adam adam 4096 Okt 23  2020 some.df
drwxr-xr-x 2 adam adam 4096 Sep  1  2020 Templates
-rw-rw-r-- 1 adam adam  0 Jan 24  2022 temp.mbsync
drwxr-xr-x 3 adam adam 4096 Mai 16 16:07 Videos
drwxrwxr-x 3 adam adam 4096 Jan 25  2019 virtual
drwxrwxr-x 5 adam adam 4096 Mai 30 10:38 'VirtualBox VMs'
-rw-rw-r-- 1 adam adam 154 Mär 16 13:11 zen.json
drwxr-xr-x 9 adam adam 4096 Aug 18 22:36 Zotero
adam@posidonia:~$
```

cd_<path_to_directory>

Go to a directory

- Can be relative or absolute!

```
adam@posidonia: ~/my_dir 73x30
adam@posidonia:~$ cd my_dir
adam@posidonia:~/my_dir$
```

```
adam@posidonia: ~/my_dir 73x30
adam@posidonia:~$ cd /home/adam/my_dir
adam@posidonia:~/my_dir$
```

`cd_ . .`

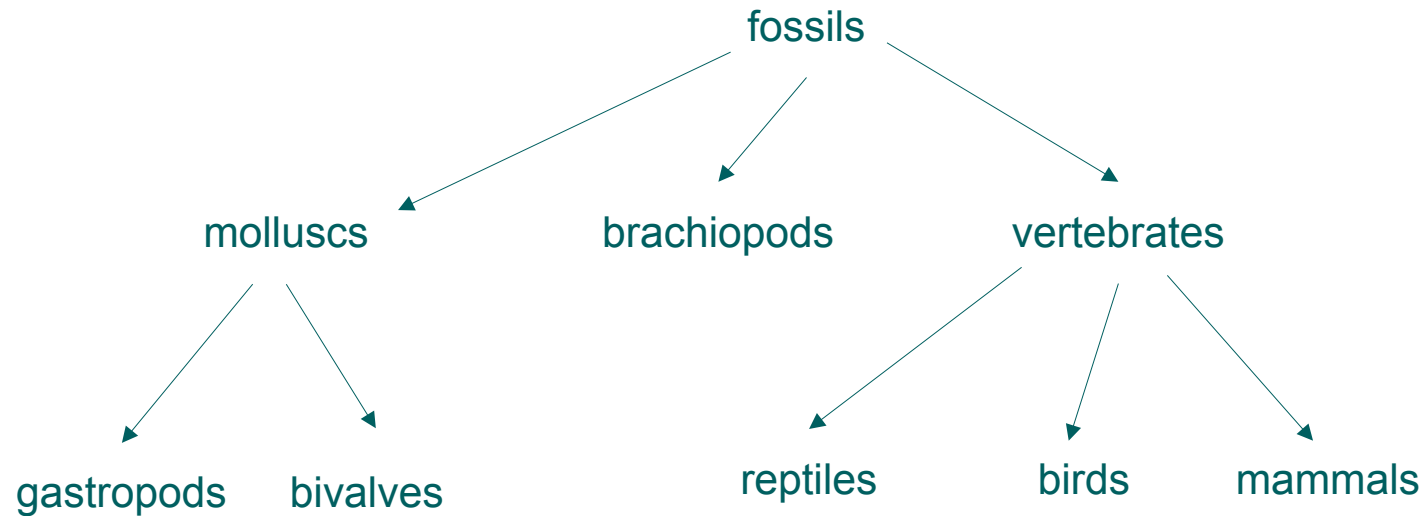
Go to parent directory

- .. (dot dot) is a placeholder for the parent of the current directory (one up in the hierarchy)

```
adam@posidonia: ~ 73x30
adam@posidonia:~$ cd /home/adam/my_dir
adam@posidonia:~/my_dir$ cd ..
adam@posidonia:~$
```

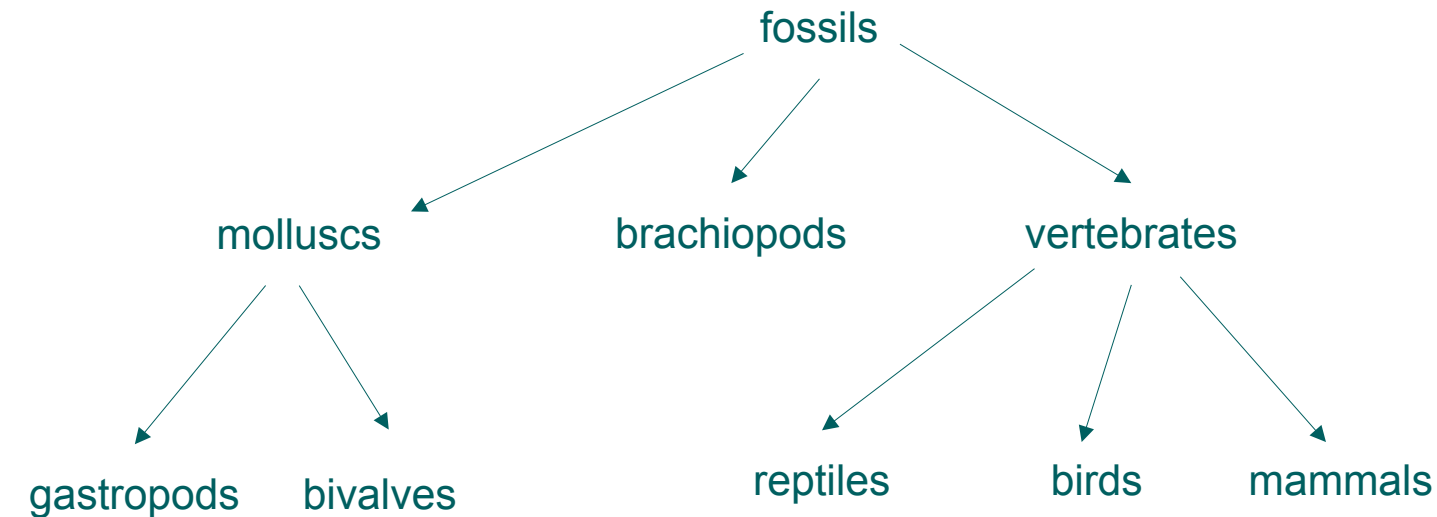
Exercise (5 minutes)

- Create this directory structure using the combinations of the previous commands!



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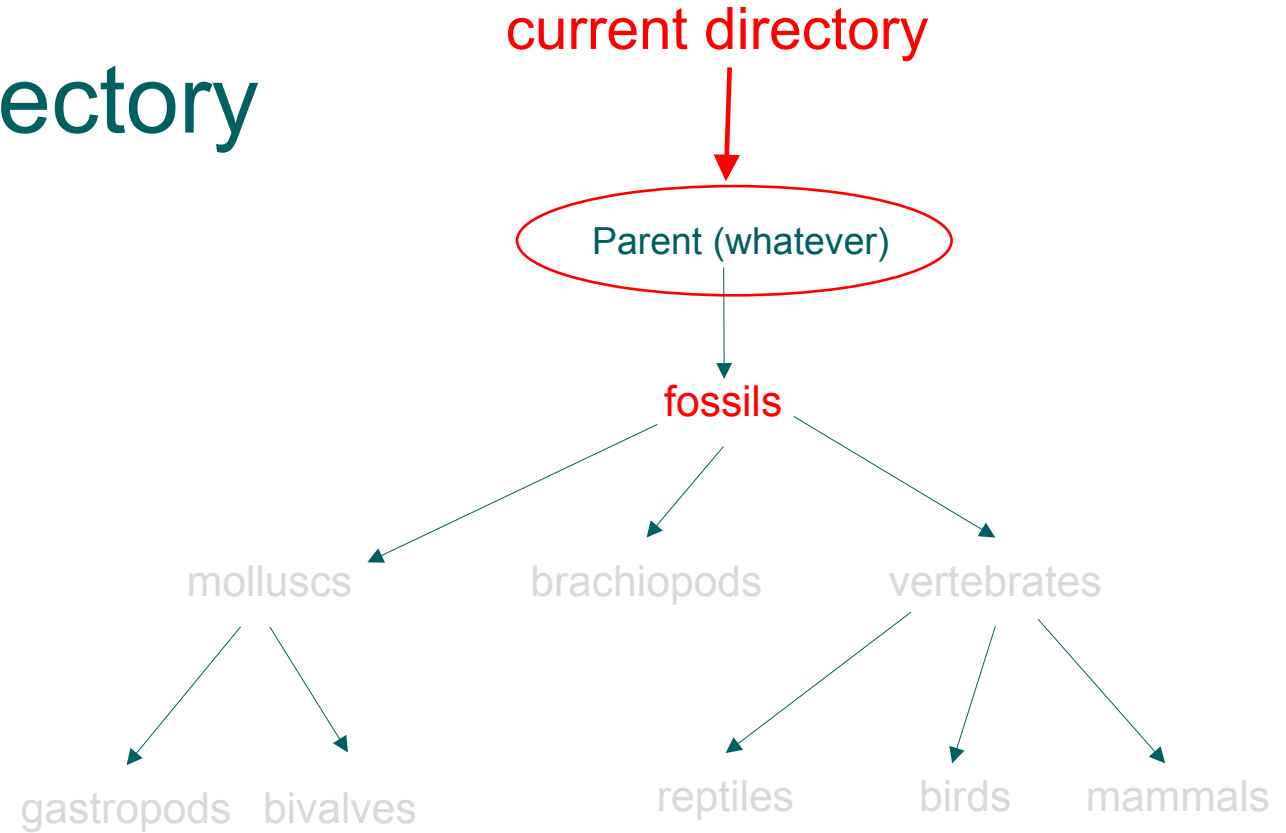


Hints

- use <TAB> completion (try <TAB> <TAB> to see multiple solutions)
- use <UP> and <DOWN> to search command history for already given commands

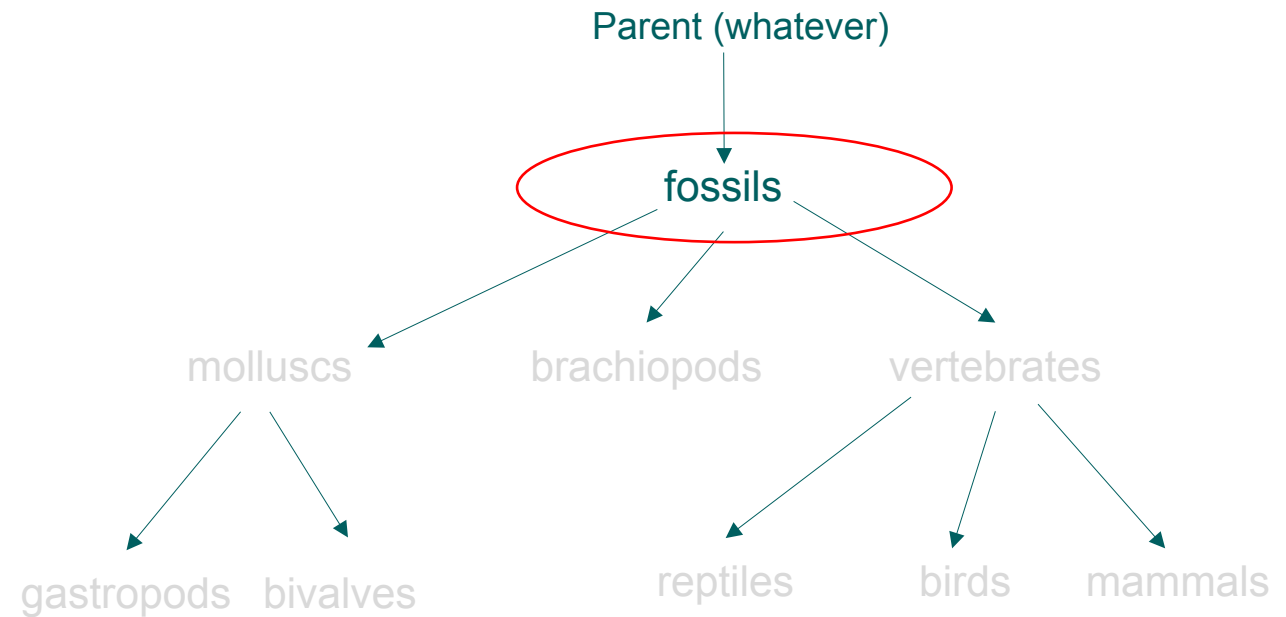
Solution 1 – changing directory

```
adam@posidonia: ~/fossils/vertebrates 73x30
adam@posidonia:~$ mkdir fossils
adam@posidonia:~$ cd fossils
adam@posidonia:~/fossils$ mkdir molluscs
adam@posidonia:~/fossils$ mkdir brachiopods vertebrates
adam@posidonia:~/fossils$ cd molluscs
adam@posidonia:~/fossils/molluscs$ mkdir gastropods bivalves
adam@posidonia:~/fossils/molluscs$ cd ..
adam@posidonia:~/fossils$ cd vertebrates
adam@posidonia:~/fossils/vertebrates$ mkdir reptiles birds mammals
adam@posidonia:~/fossils/vertebrates$
```



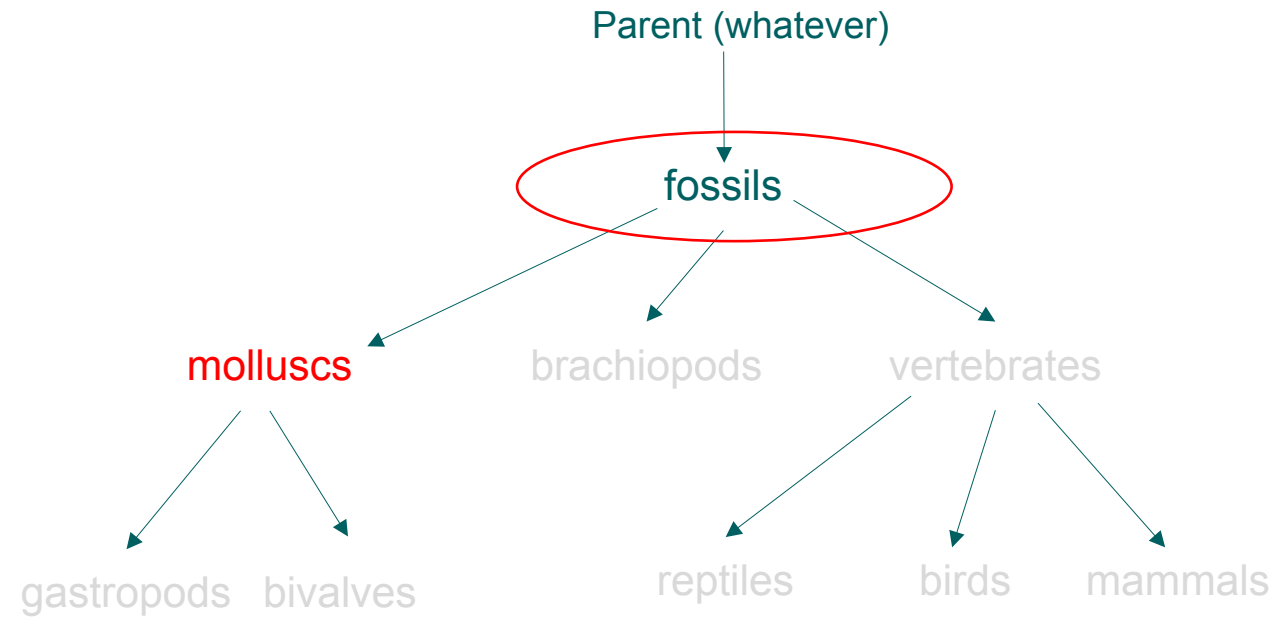
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adam@posidonia:~/fossils/molluscs$ cd ..
adam@posidonia:~/fossils$ cd vertebrates
adam@posidonia:~/fossils/vertebrates$ mkdir reptiles birds mammals
adam@posidonia:~/fossils/vertebrates$
```



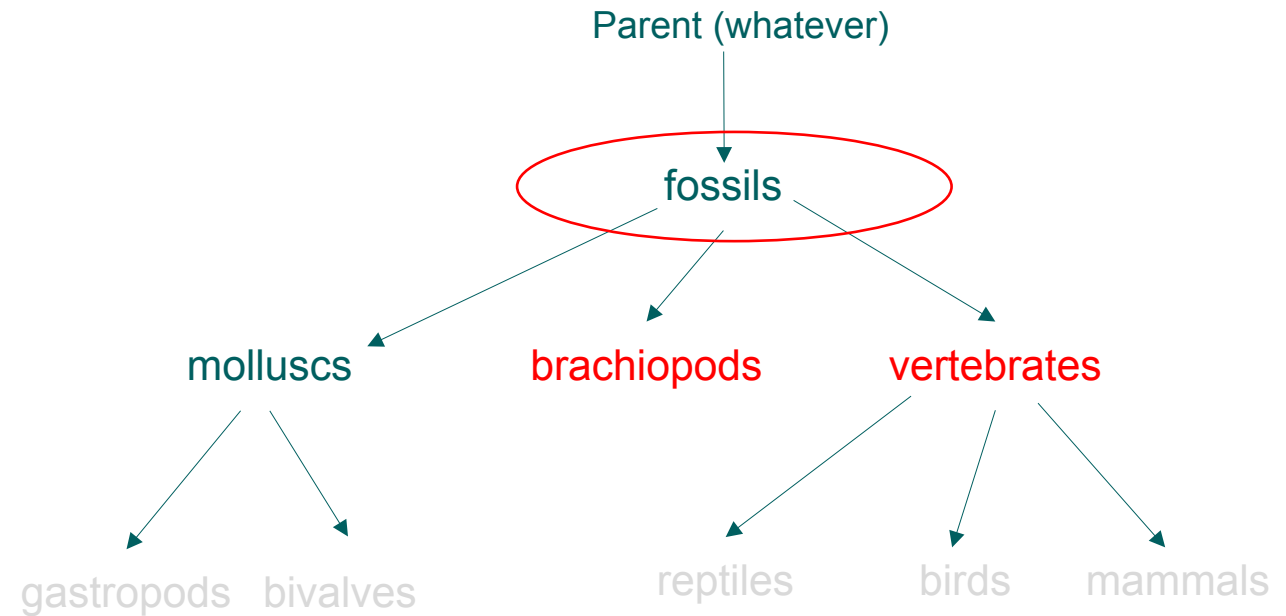
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adam@posidonia:~/fossils$ cd vertebrates
adam@posidonia:~/fossils/vertebrates$ mkdir reptiles birds mammals
adam@posidonia:~/fossils/vertebrates$
```



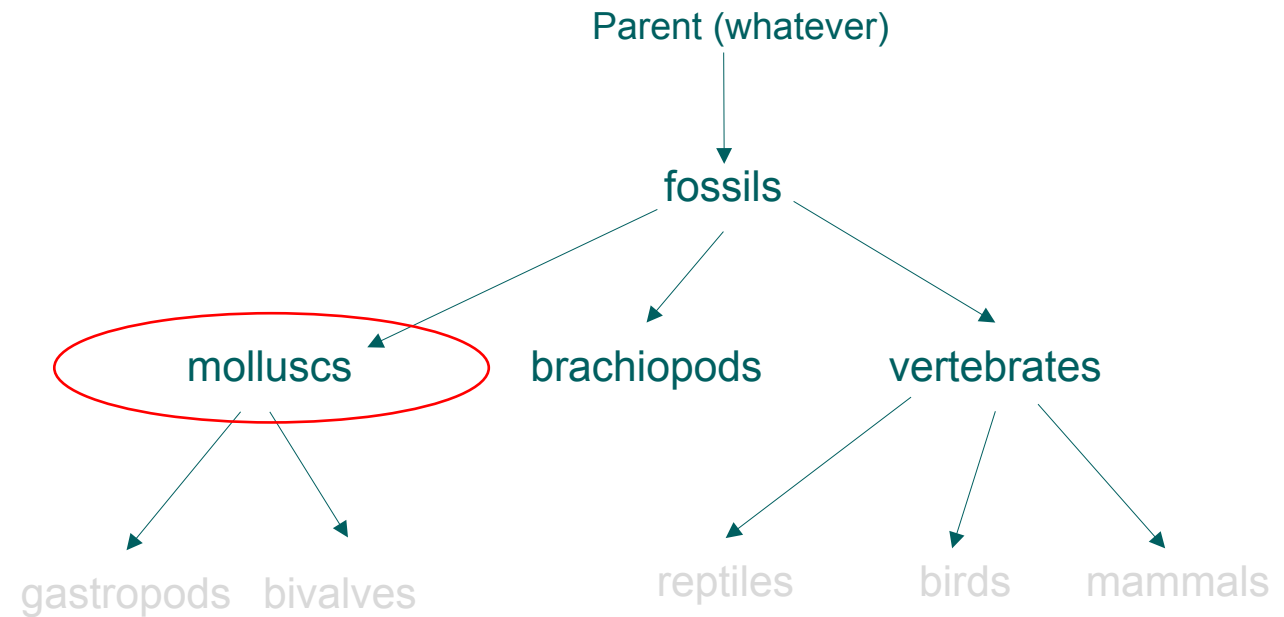
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adam@posidonia:~/fossils/vertebrates$ mkdir reptiles birds mammals
adam@posidonia:~/fossils/vertebrates$
```



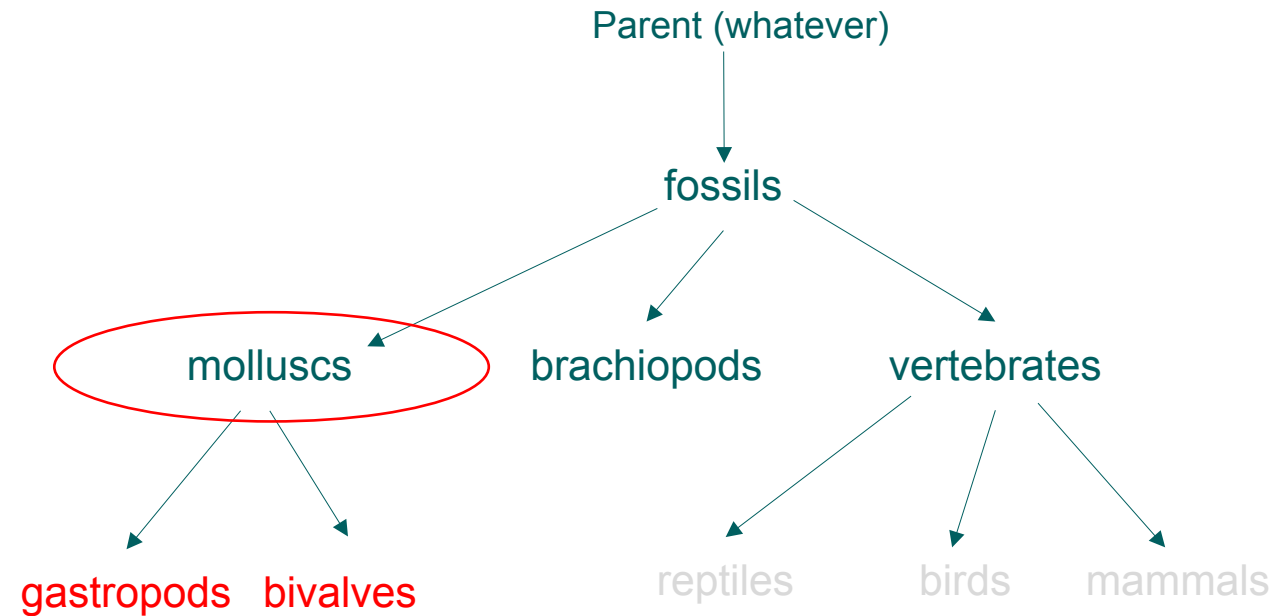
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```



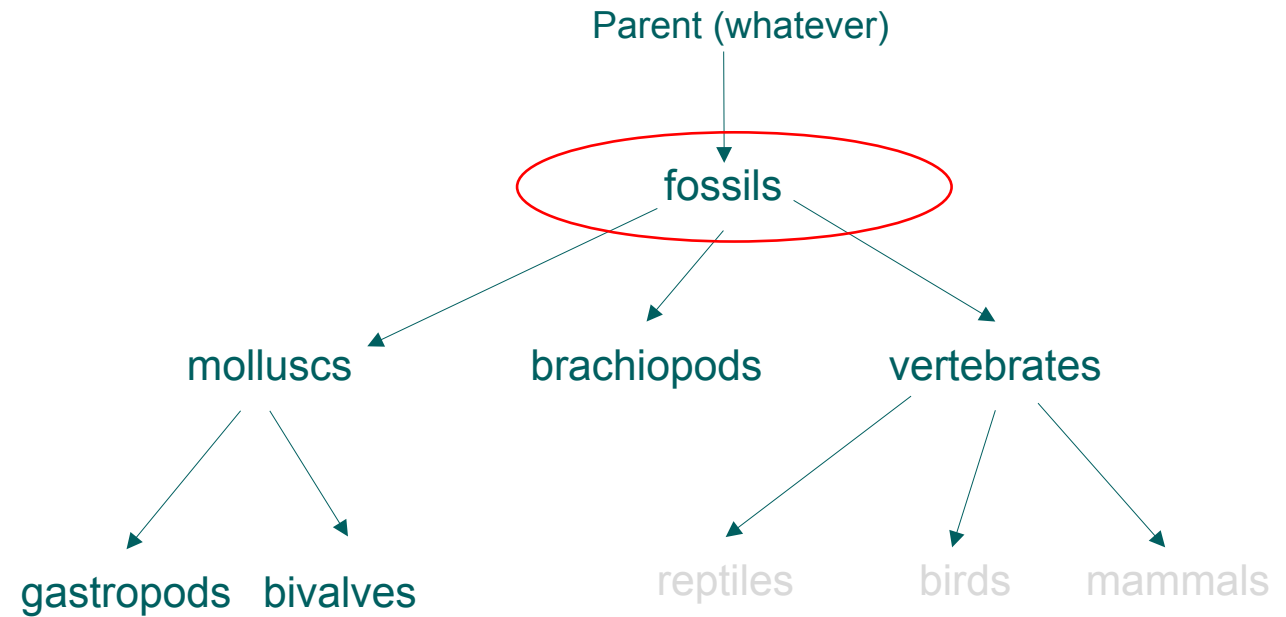
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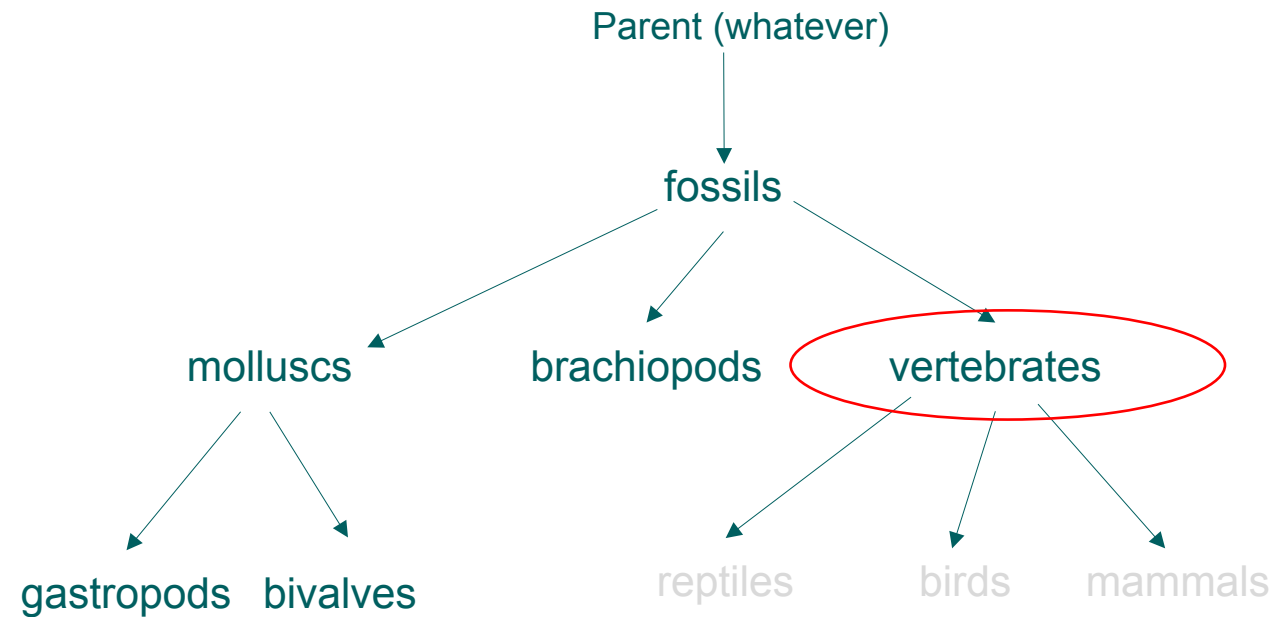
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```



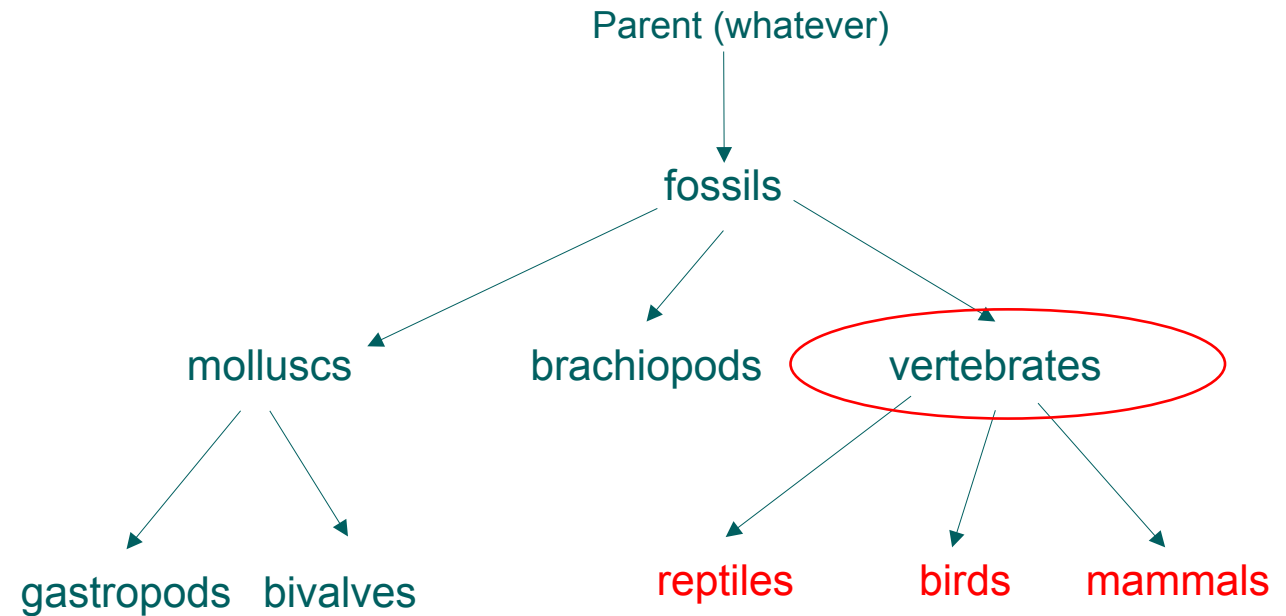
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```



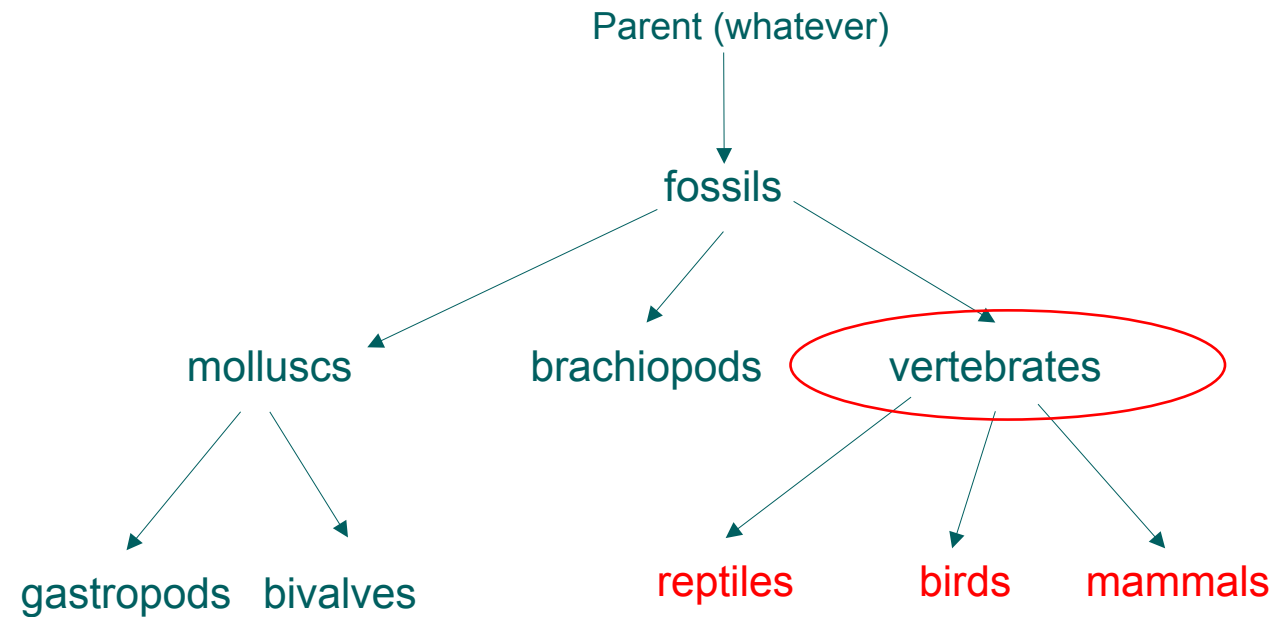
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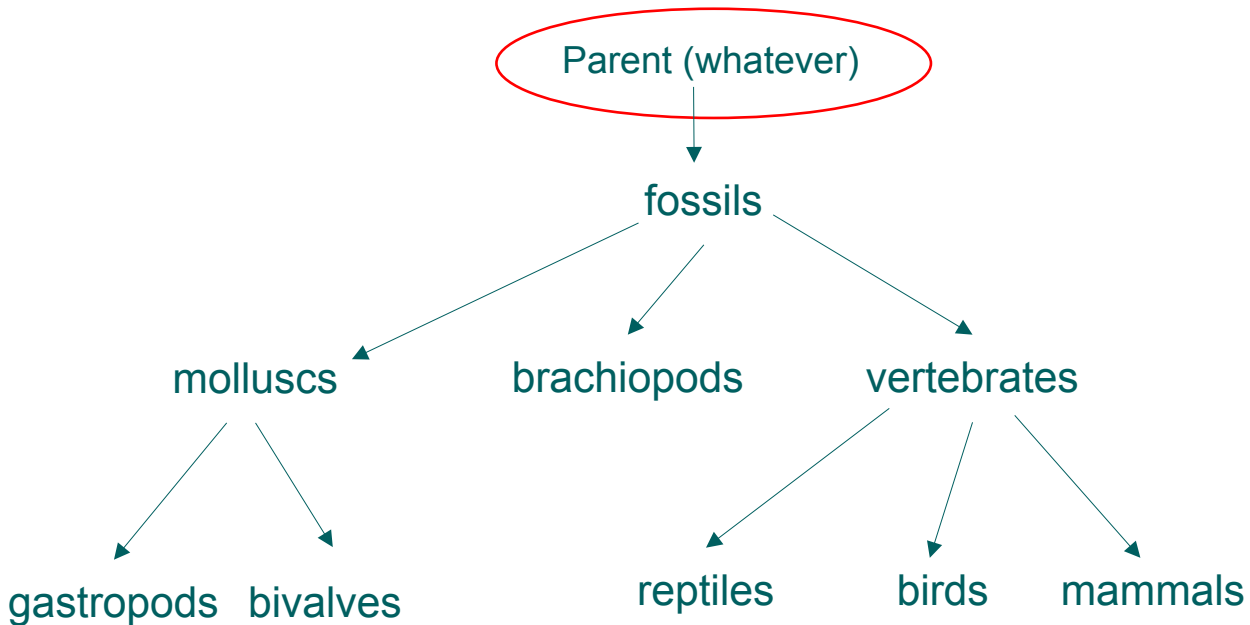


How to check?

find_<path to directory>

Recursive listing

- 1. Go back to the parent
- 2. Use find there!



```
adam@posidonia: ~ 73x30
adam@posidonia:~/fossils/vertebrates$ cd ../../
adam@posidonia:~$ find fossils
fossils
fossils/molluscs
fossils/molluscs/gastropos
fossils/molluscs/bivalves
fossils/brachiopods
fossils/vertebrates
fossils/vertebrates/reptiles
fossils/vertebrates/birds
fossils/vertebrates/mammals
adam@posidonia:~$
```

parent of parent

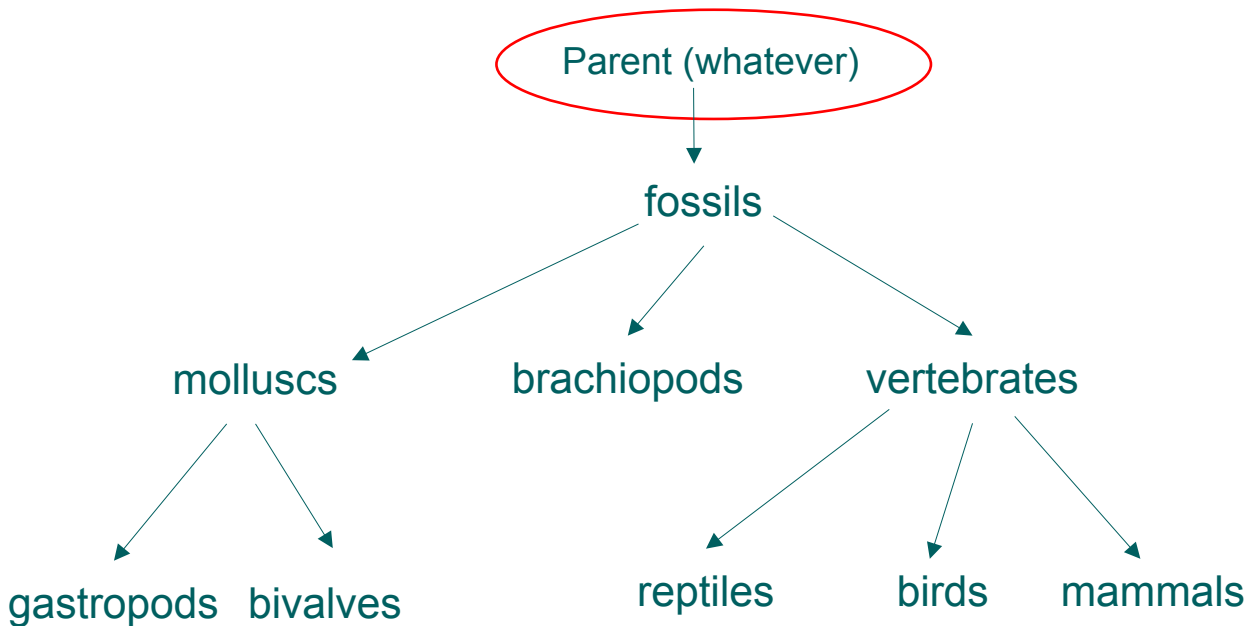
relative paths

`find` **<path>** **>** **<path_to file>**

Angled bracket or *chevron*

Output redirection

- Whatever was output to the console is now in a new file!



```
adam@posidonia: ~ 73x30
adam@posidonia:~/fossils/vertebrates$ cd ../../
adam@posidonia:~$ find fossils
fossils
fossils/molluscs
fossils/molluscs/gastropods
fossils/molluscs/bivalves
fossils/brachiopods
fossils/vertebrates
fossils/vertebrates/reptiles
fossils/vertebrates/birds
fossils/vertebrates/mammals
adam@posidonia:~$ find fossils > fossil_path.txt
adam@posidonia:~$
```

">" Will overwrite existing files!

Suggested nomenclature

() : **Parenthesis** (open and close)
[] : **Bracket** (open and close)
{ } : **Brace** (open and close)
< > : **Chevrons** (left and right)

`cat_<path to file>`

Display contents of file

- Exactly as it was output to the console

```
adam@posidonia: ~ 73x30
adam@posidonia:~$ cat fossil_path.txt
fossils
fossils/molluscs
fossils/molluscs/gastropods
fossils/molluscs/bivalves
fossils/brachiopods
fossils/vertebrates
fossils/vertebrates/reptiles
fossils/vertebrates/birds
fossils/vertebrates/mammals
adam@posidonia:~$
```

`rm -r <path to dir>`

Recursive deletion (-r)

- Deletes the content of the directory and the directory itself
- `rmdir` doesn't work!
- No output = success?!

```
adam@posidonia: ~ 73x30
adam@posidonia:~$ cat fossil_path.txt
fossils
fossils/molluscs
fossils/molluscs/gastropods
fossils/molluscs/bivalves
fossils/brachiopods
fossils/vertebrates
fossils/vertebrates/reptiles
fossils/vertebrates/birds
fossils/vertebrates/mammals
adam@posidonia:~$ rm -r fossils ←
adam@posidonia:~$
```

WARNING!
The results of `rm`
cannot be undone!

echo <text>

Print something

- Used to print things to the console (standard output)
- \$? Is a special symbol: the exit code of the last command:
 - 0: Success
 - Other: Failure

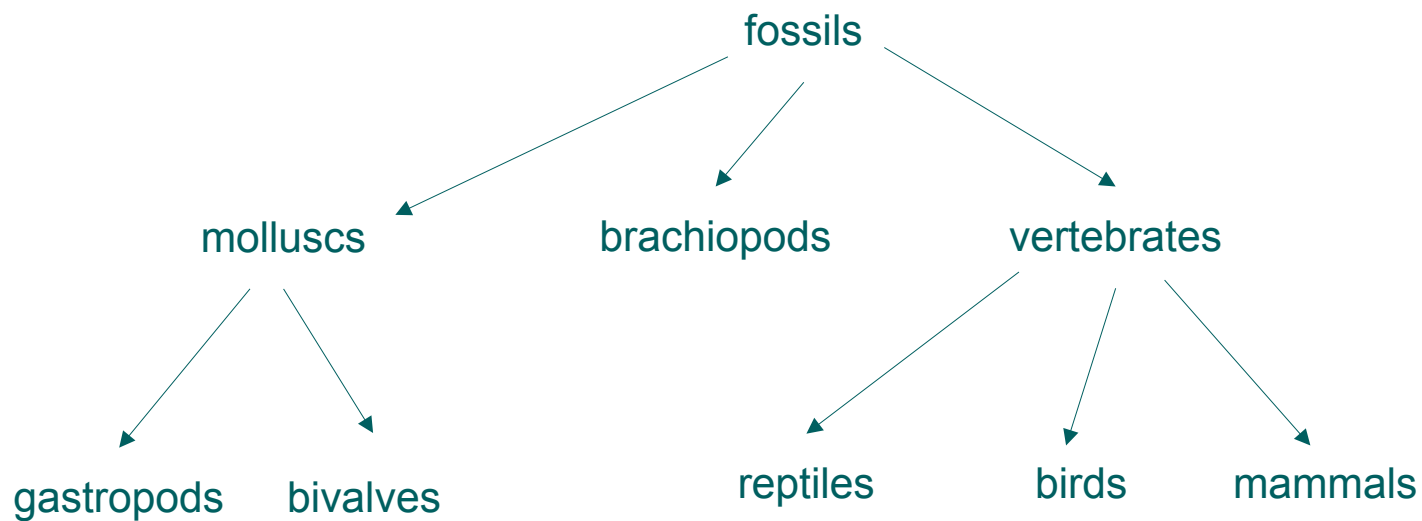
<https://www.redhat.com/sysadmin/exit-codes-demystified>

adam@posidonia: ~ 73x30

```
adam@posidonia:~$ cat fossil_path.txt
fossils
fossils/molluscs
fossils/molluscs/gastropods
fossils/molluscs/bivalves
fossils/brachiopods
fossils/vertebrates
fossils/vertebrates/reptiles
fossils/vertebrates/birds
fossils/vertebrates/mammals
adam@posidonia:~$ rm -r fossils
adam@posidonia:~$ echo $? ←
0
adam@posidonia:~$
```

Recreate the structure!

- Did you type things into the console?!



Hint 1. Use a general-purpose text editor!

Novice-friendly:

- Sublime Text



- VS Code



- Atom



Expert-friendly:

- Vim

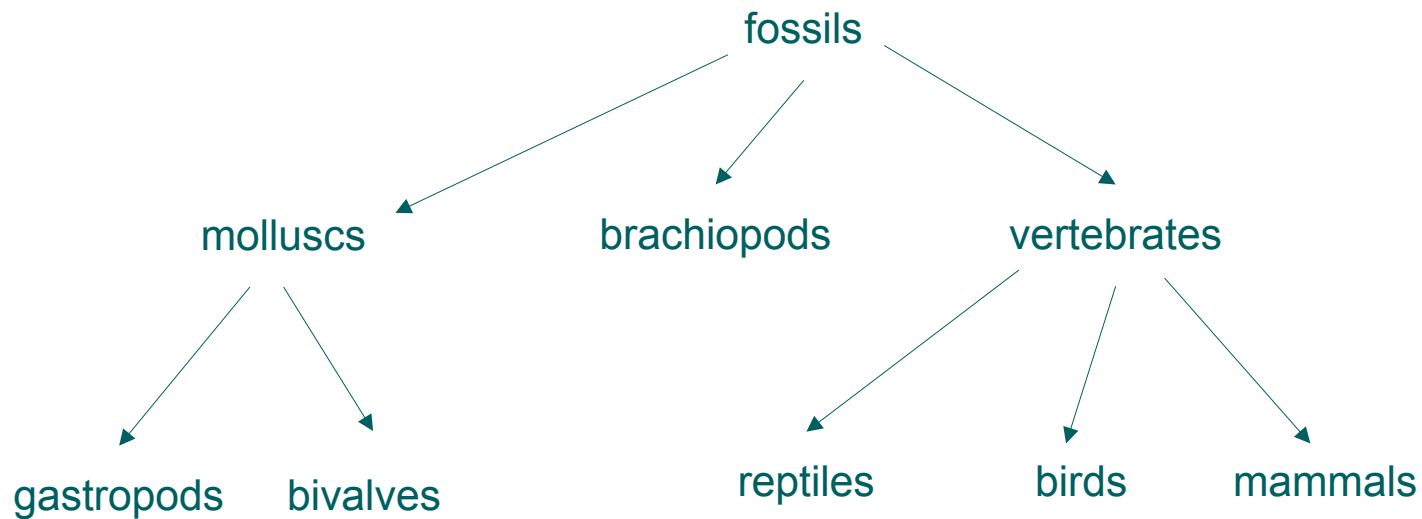


- Emacs



Solution 2

- Using the same reference directory



Hint 2. we can use the contents of fossil_path.txt

Add `mkdir` in front of every line, then copy and paste into the console!

```
1 mkdir fossils
2 mkdir fossils/molluscs
3 mkdir fossils/molluscs/gastropods
4 mkdir fossils/molluscs/bivalves
5 mkdir fossils/brachiopods
6 mkdir fossils/vertebrates
7 mkdir fossils/vertebrates/reptiles
8 mkdir fossils/vertebrates/birds
9 mkdir fossils/vertebrates/mammals
```

bash_--version

Running console applications

- `--version`: ask for program version
- `--help`: display help for program

```
adam@posidonia: ~ 73x30
adam@posidonia:~$ bash --version
bash --version
GNU bash, version 5.0.17(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2019 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
adam@posidonia:~$
```

`bash_<path>`

Executing shell scripts

- The text we created is actually a shell script draft
- The “bash” console application program can be used to execute it.

<https://www.redhat.com/sysadmin/exit-codes-demystified>

```
adam@posidonia: ~ 73x30
adam@posidonia:~$ find fossils
find: 'fossils': No such file or directory
adam@posidonia:~$ vim fossil_path.txt
adam@posidonia:~$ bash fossil_path.txt
adam@posidonia:~$ find fossils
fossils
fossils/molluscs
fossils/molluscs/gastropods
fossils/molluscs/bivalves
fossils/brachiopods
fossils/vertebrates
fossils/vertebrates/reptiles
fossils/vertebrates/birds
fossils/vertebrates/mammals
adam@posidonia:~$
```

Annotations:

- directory not present (points to the error message)
- adding 'mkdir' to previous file (points to the `vim fossil_path.txt` command)
- execute file as bash script (points to the `bash fossil_path.txt` command)
- Show results (points to the output of the `find` command)

`chmod_+x_<path>`

Making executables

- Works with any file!
- Can be used to change file attributes: read, write, execute (for every user)
- “Shebang” indicates the kind of language used to interpret the file, for bash: “#!/bin/bash”
- Typical shell script files with .sh
- Execution requires a path!

Shebang

```
1 #!/bin/bash
2
3 mkdir fossils
4 mkdir fossils/molluscs
5 mkdir fossils/brachiopods
6 mkdir fossils/vertebrates
7 mkdir fossils/vertebrates/reptiles
8 mkdir fossils/vertebrates/birds
9 mkdir fossils/vertebrates/mammals
10 mkdir fossils/molluscs/bivalves
11 mkdir fossils/molluscs/gastropods
12
13 echo "You have run the fossils.sh file!"
```

```
adam@TheStuff ~/work$ ls -l
total 4
-rw-rw-r-- 1 adam adam 311 aug 18 05:26 fossils.sh
adam@TheStuff ~/work$ chmod +x fossils.sh
adam@TheStuff ~/work$ ls -l
total 4
-rwxrwxr-x 1 adam adam 311 aug 18 05:26 fossils.sh
adam@TheStuff ~/work$ ./fossils.sh
You have run the fossils.sh file!
adam@TheStuff ~/work$ find .
.
./fossils.sh
./fossils
./fossils/brachiopods
./fossils/molluscs
./fossils/molluscs/gastropods
./fossils/molluscs/bivalves
./fossils/vertebrates
./fossils/vertebrates/reptiles
./fossils/vertebrates/mammals
./fossils/vertebrates/birds
adam@TheStuff ~/work$ |
```

Not executable

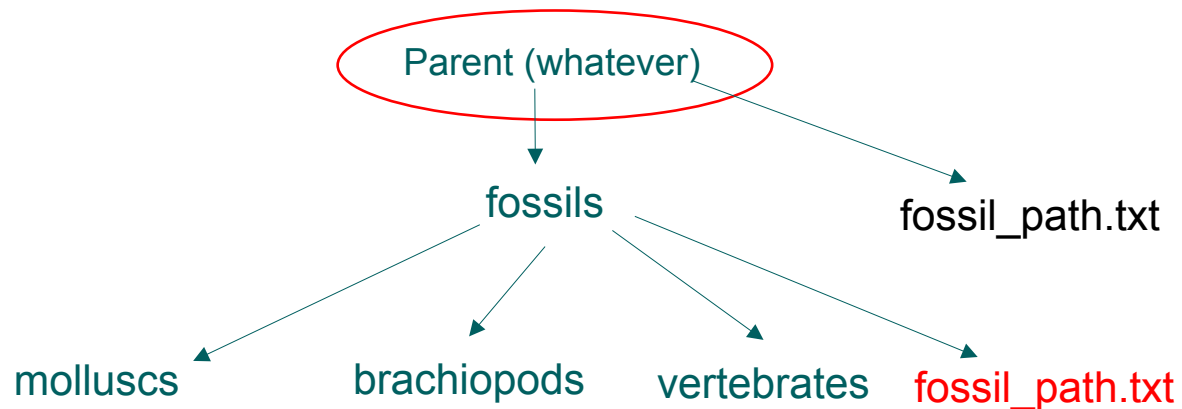
Executable

Execution

`cp_<what>_<where>`

Copying a file or directory

- Target directory or file
- If directory, the file will be put into it



```
adam@posidonia: ~ 73x30
adam@posidonia:~$ cp fossil_path.txt fossils
adam@posidonia:~$ ls fossils
brachiopods fossil_path.txt molluscs vertebrates
adam@posidonia:~$
```

List contents of path!

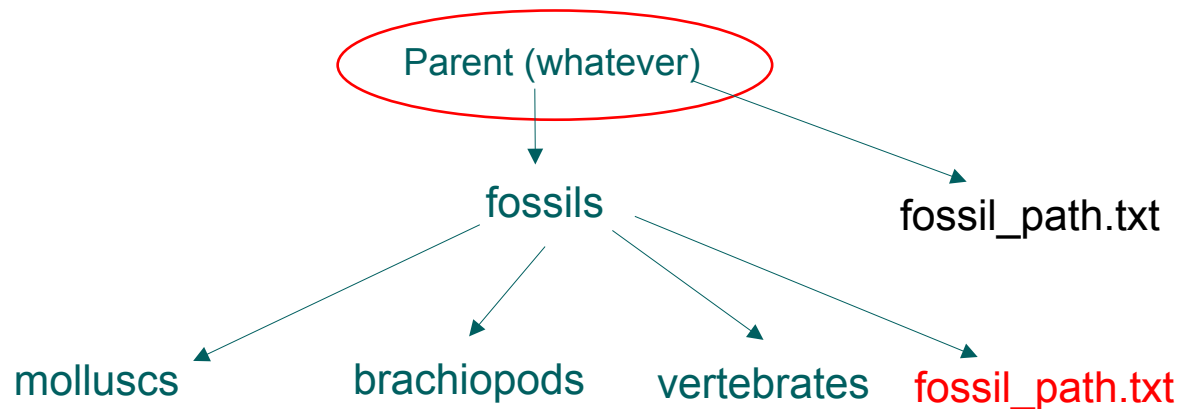
New file

One of the tools that we looked at can be used to delete the file that we have created. Try to delete it!

`rm_<path_file>`

Without `-r` removes a single file

- As with `cp`, multiple files can be passed to this (separated by spaces)

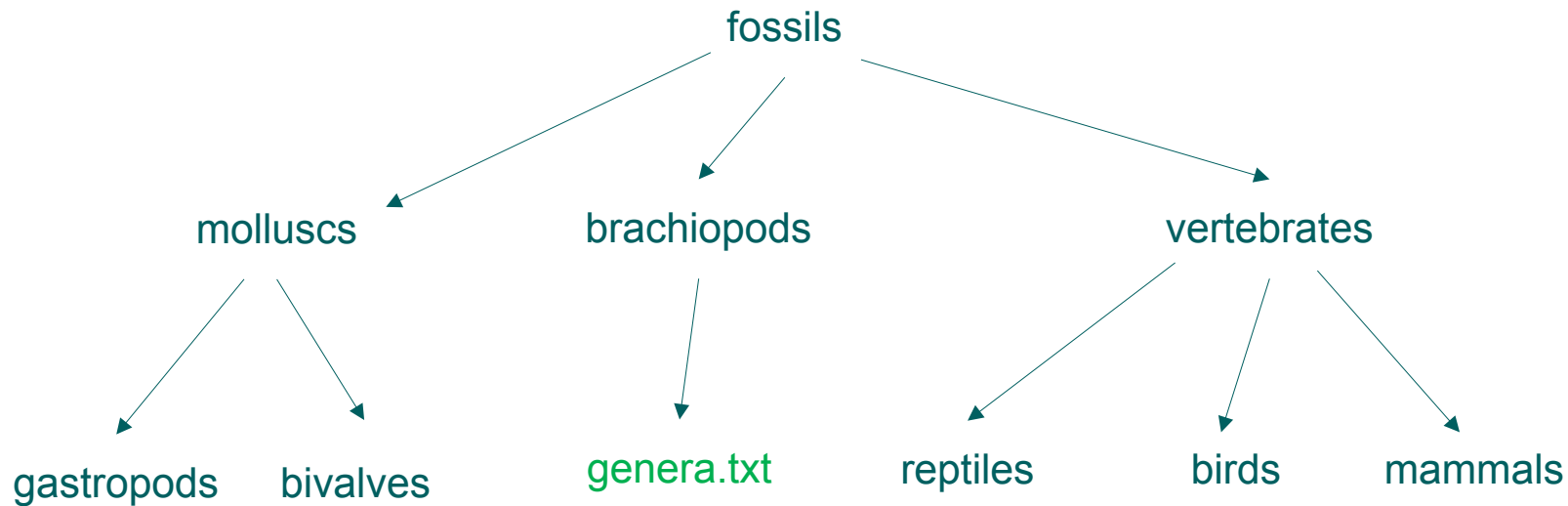


```
adam@posidonia: ~ 73x30
adam@posidonia:~$ cp fossil_path.txt fossils
adam@posidonia:~$ ls fossils
brachiopods fossil_path.txt molluscs vertebrates
adam@posidonia:~$ rm fossils/fossil_path.txt
adam@posidonia:~$ ls fossils
brachiopods molluscs vertebrates
adam@posidonia:~$
```

File disappeared

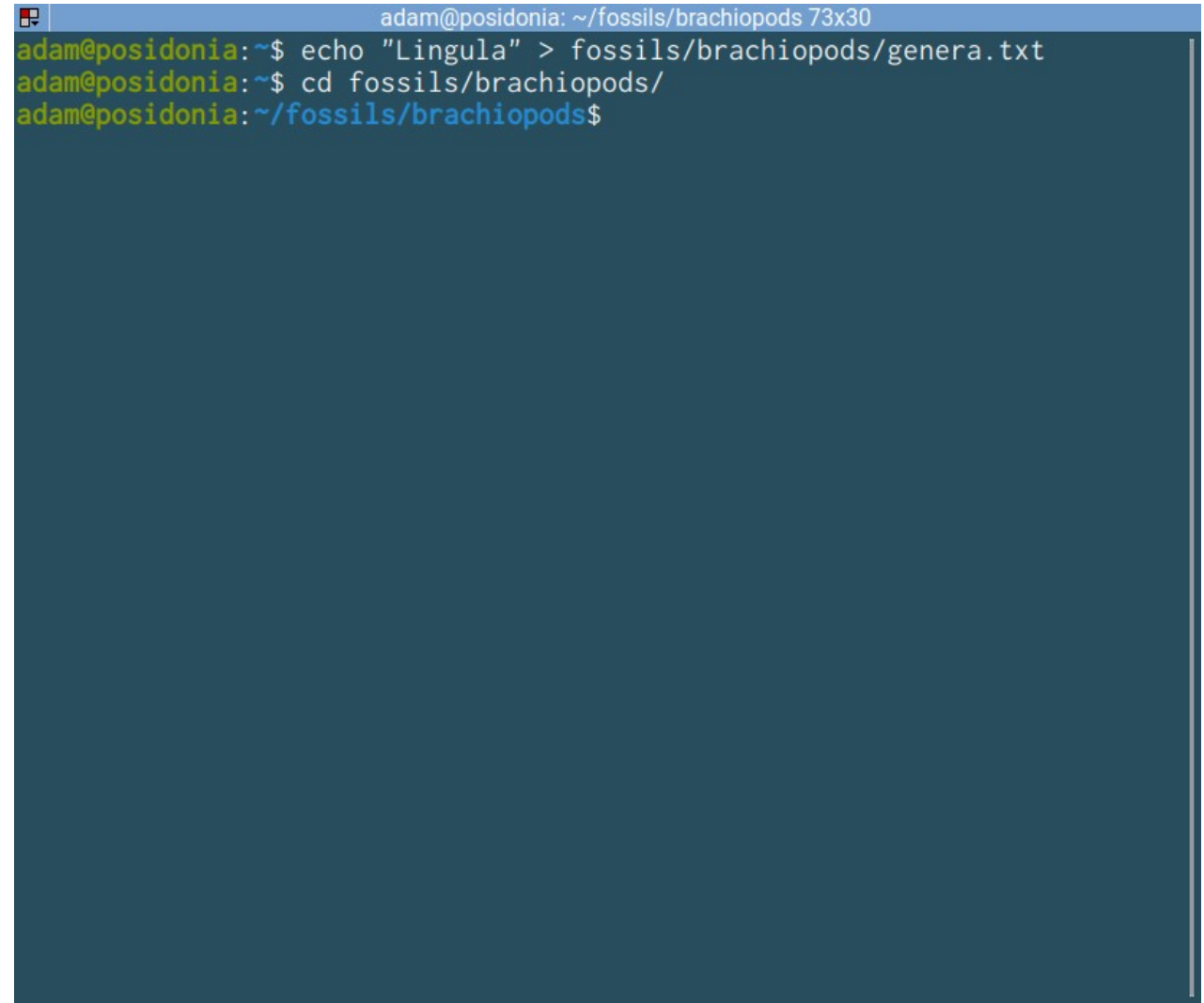
Demo!

1. Use an `echo` statement to write the genus name “*Lingula*” into `fossils/brachiopods/genera.txt`!
2. Then change directory to `brachiopods`.



```
echo "Lingula" > fossils/brachiopods/genera.txt
```

- You can use the double chevron
>> to append to an existing file



```
adam@posidonia: ~/fossils/brachiopods 73x30
adam@posidonia:~$ echo "Lingula" > fossils/brachiopods/genera.txt
adam@posidonia:~$ cd fossils/brachiopods/
adam@posidonia:~/fossils/brachiopods$
```

`echo "Spiriferina" >> genera.txt`

Appending to files

- You can use the double chevron >> to append to an existing file
- Added to new line!

```
adam@posidonia: ~/fossils/brachiopods 73x30
adam@posidonia:~$ echo "Lingula" > fossils/brachiopods/genera.txt
adam@posidonia:~$ cd fossils/brachiopods/
adam@posidonia:~/fossils/brachiopods$ echo "Spiriferina" >> genera.txt
adam@posidonia:~/fossils/brachiopods$ cat genera.txt
Lingula
Spiriferina
adam@posidonia:~/fossils/brachiopods$
```

Special characters

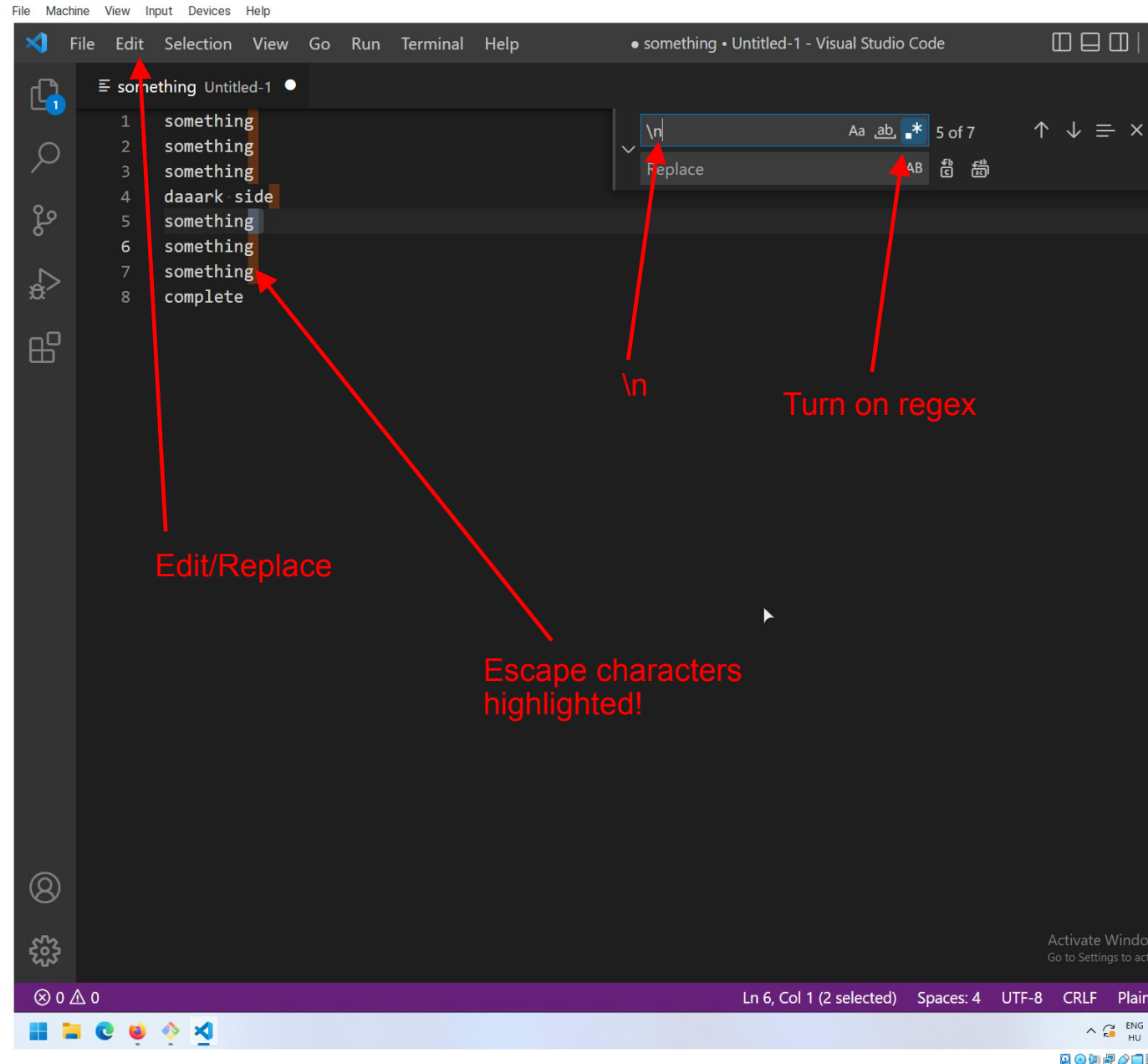
We use so called “escape characters” to denote special symbols, that sometimes have other meanings.

`\n`: newline escape

`\t`: tab escape

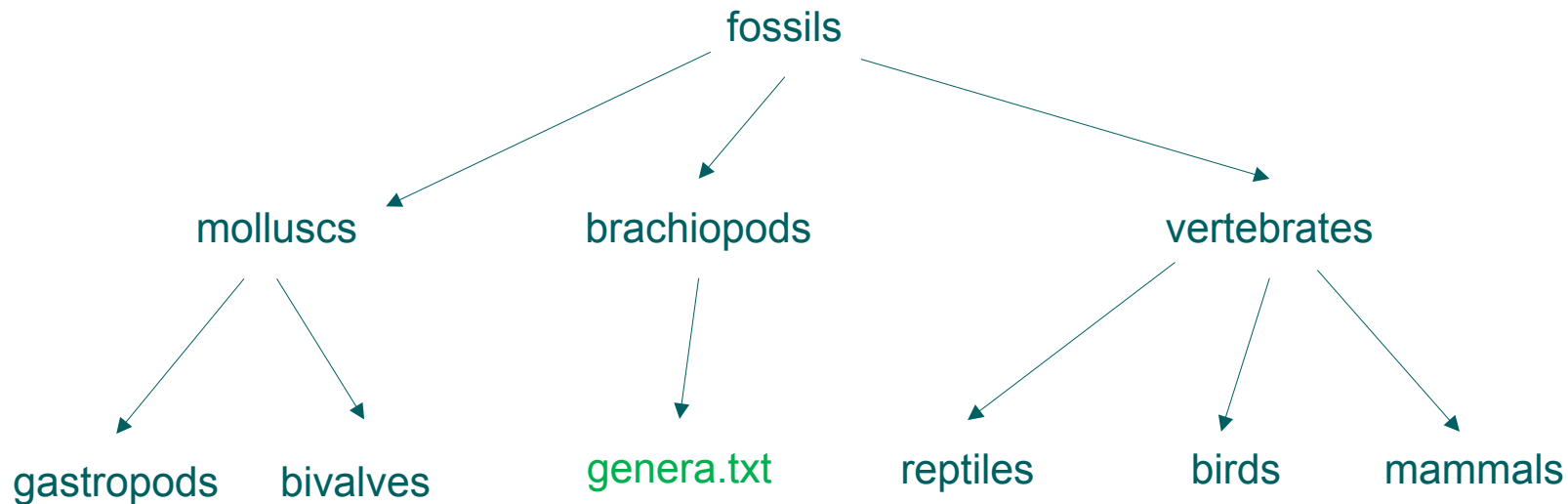
`\` : double quote escape

`'` : single quote escape



Demo!

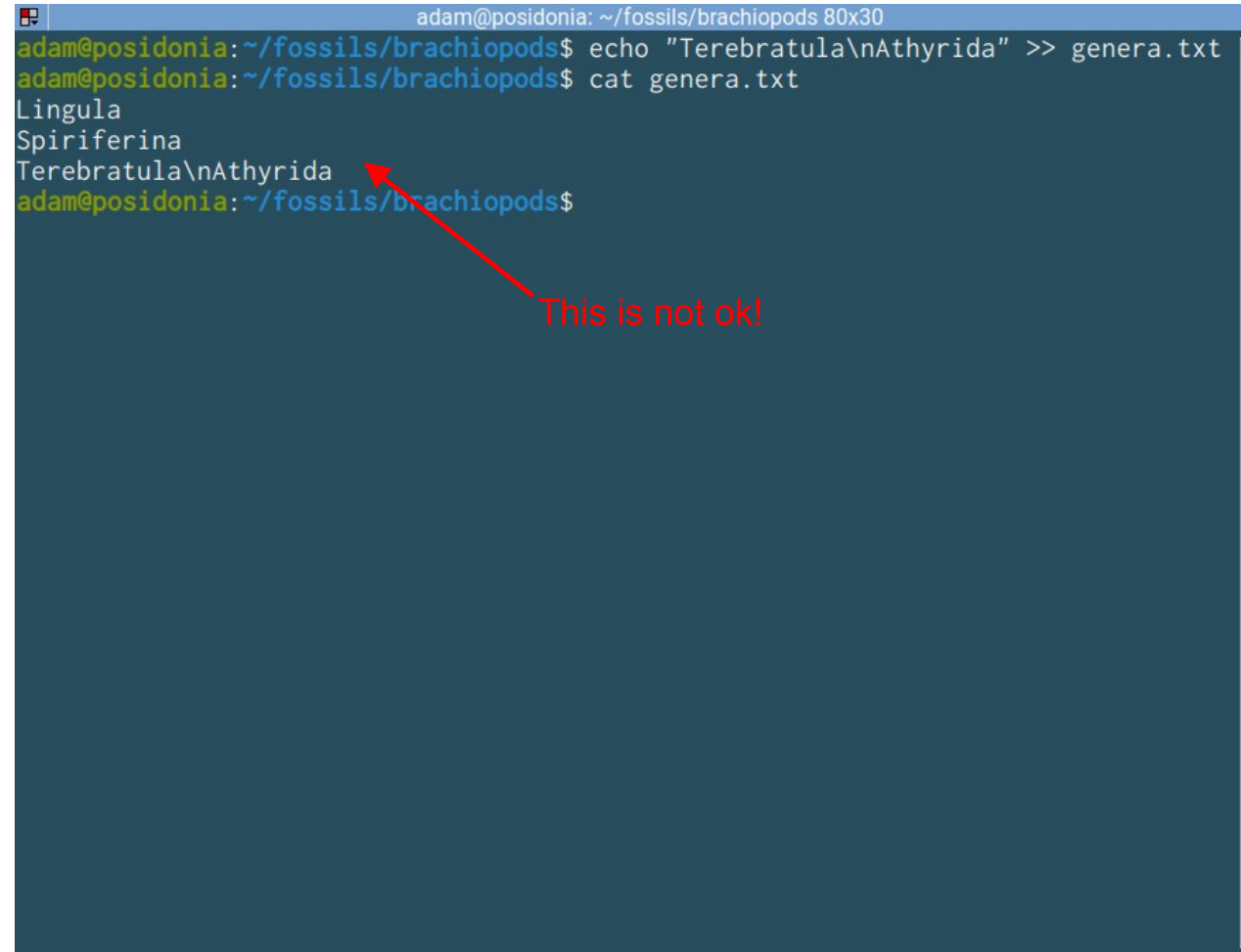
1. Use an `echo` statement to also add the genus name “*Terebratula*” and “*Athyris*” into `fossils/brachiopods/genera.txt`, **use a newline escape between them!**
2. Then change directory to `brachiopods`.



```
echo "Terebratula\nAthyrida" >> genera.txt
```

Appending to files

- It doesn't seem to work!
- Echo needs to know to replace the combination `\n` with the newline character!

A terminal window with a dark blue background and light blue title bar. The title bar text is 'adam@posidonia: ~/fossils/brachiopods 80x30'. The terminal shows the following commands and output:

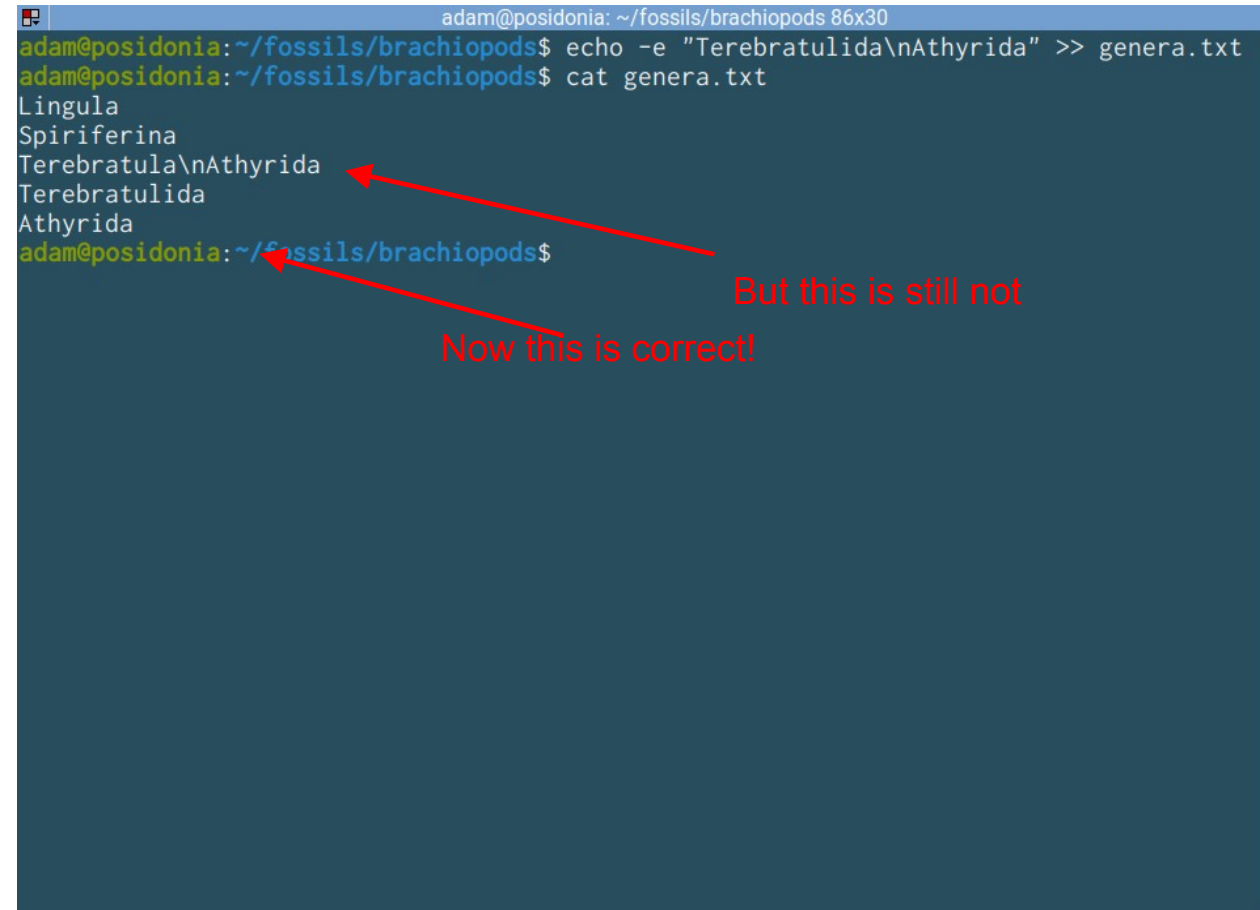
```
adam@posidonia:~/fossils/brachiopods$ echo "Terebratula\nAthyrida" >> genera.txt
adam@posidonia:~/fossils/brachiopods$ cat genera.txt
Lingula
Spiriferina
Terebratula\nAthyrida
adam@posidonia:~/fossils/brachiopods$
```

A red arrow points from the text 'This is not ok!' to the output line 'Terebratula\nAthyrida'.

```
echo -e "Terebratula\nAthyrida" >> genera.txt
```

Appending to files

- Use the `-e` option!
- Our file is messed up. Options:
 - 1. Redo our file
 - 2. Use an editor to correct
- Delete the bad line!
- Better, next time: go back in time.



The terminal window shows the following commands and output:

```
adam@posidonia: ~/fossils/brachiopods 86x30
adam@posidonia:~/fossils/brachiopods$ echo -e "Terebratulida\nAthyrida" >> genera.txt
adam@posidonia:~/fossils/brachiopods$ cat genera.txt
Lingula
Spiriferina
Terebratula\nAthyrida
Terebratulida
Athyrida
adam@posidonia:~/fossils/brachiopods$
```

Two red arrows point from text annotations to the output. The first arrow points to the line `Terebratula\nAthyrida` with the text "But this is still not". The second arrow points to the line `Athyrida` with the text "Now this is correct!".

Basic version control with Git

and GitHub

Why version control?

Projects evolve in a non-linear way, especially programming projects.

- Multiple people work on them, sometimes at the same time
- Recording the history of project development
- Working with many files
- Sharing code is necessary, we also need to know who changes what



Difference between Git and GitHub?

git

- Locally running application
- Operates with files in a local directory (repository)
- Works without a remotes!



GitHub and GitLab

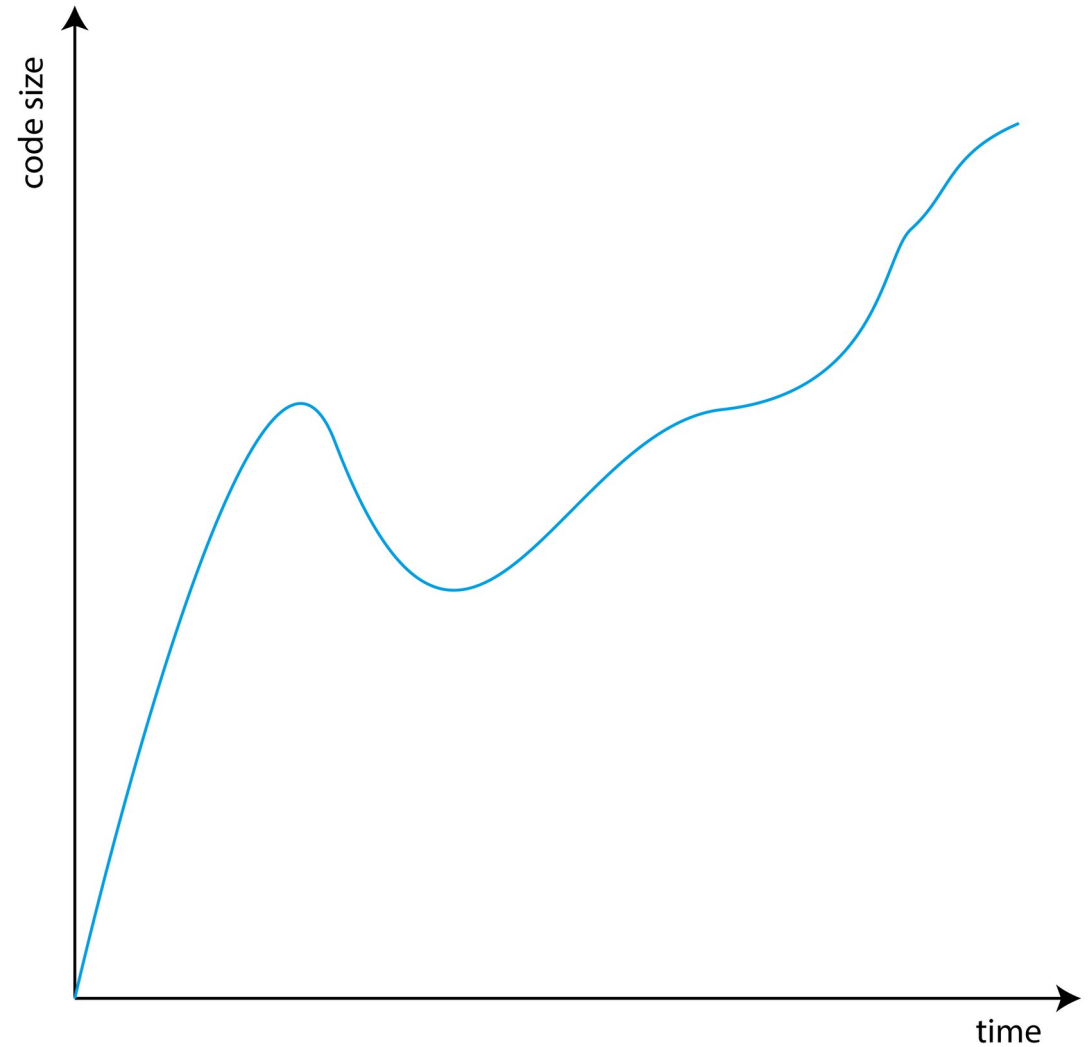
- Remote servers with copies of the repository



The basic use of git

Record snapshots of how a project develops.

- Code develops in a non-linear, but continuous way, with lots of small changes:
 - Contents of files change
 - New files are added to the repository
 - Old files are delete from the repository

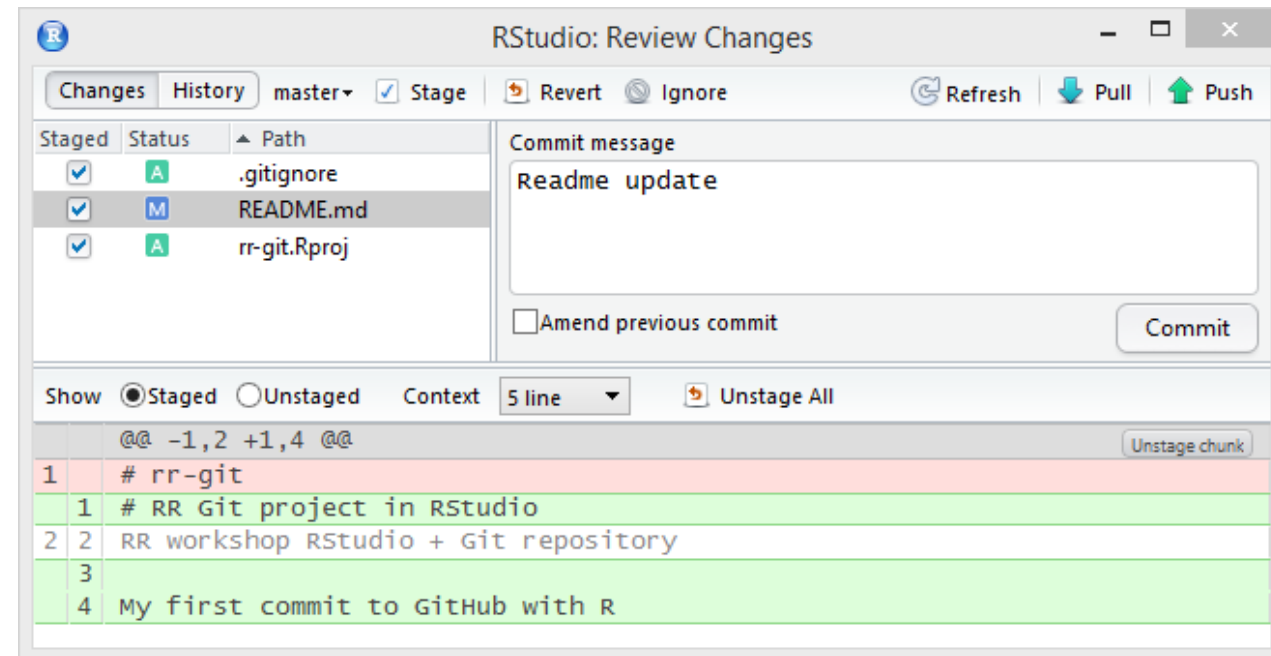


Interfaces to git

**Git is a command line
(console application)**

- The complete features are only available via the command line!
- Simplified graphical interfaces written for novices, embedded in IDEs
- These actually just translate the actions to the command line application -> Experiment!

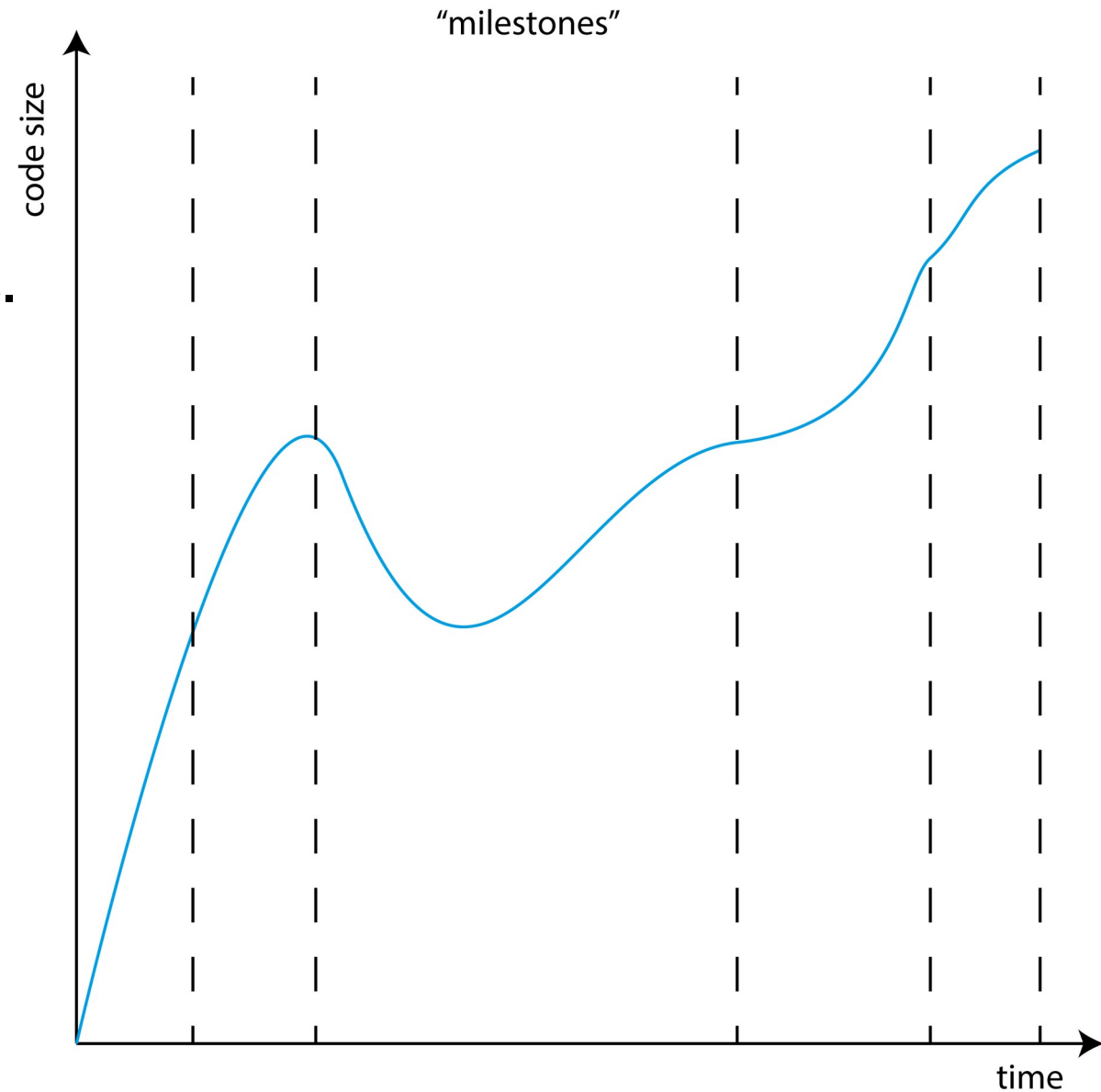
```
adam@posidonia: ~ 86x30
adam@posidonia:~$ git --version
git version 2.25.1
adam@posidonia:~$
```



The basic use of git

Record snapshots of how a project develops.

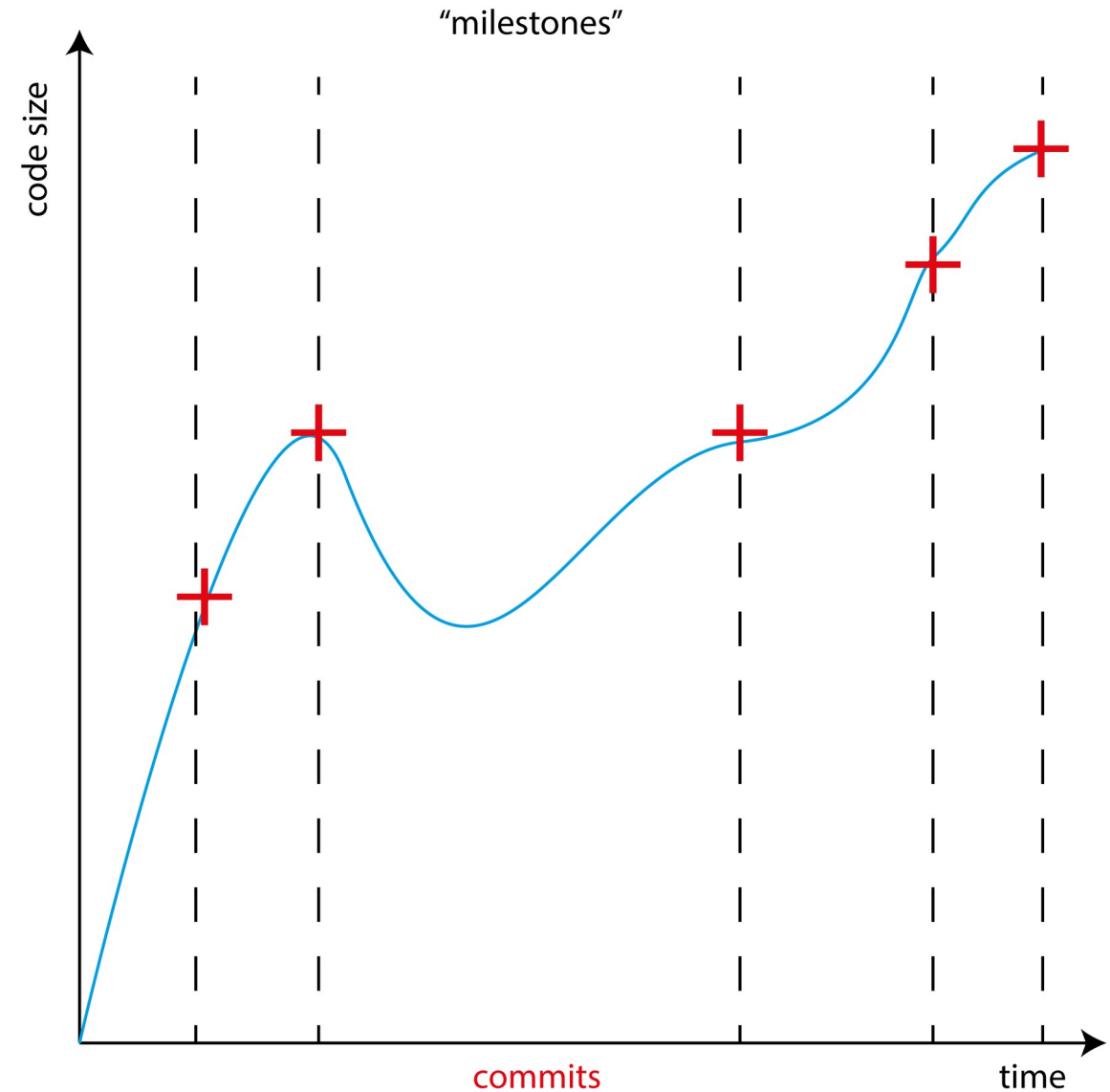
- Specific states of the code represent milestones:
 - Something works completely
 - Everything is cleaned up
 - Ready for further development
- In between these are transient states, when you are working on something but that is not yet done.



The basic use of git

Record snapshots of how a project develops.

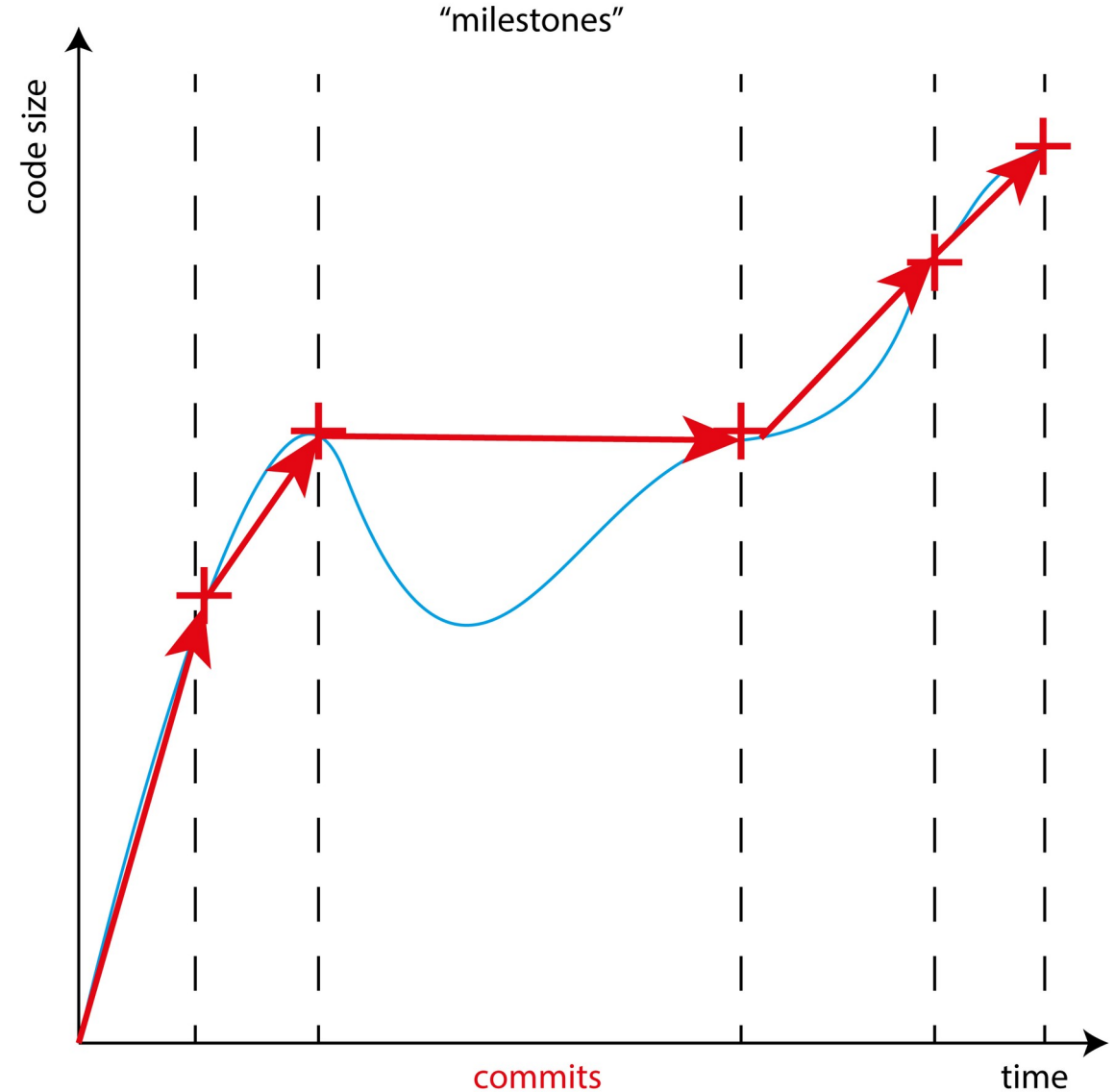
- These milestones can be saved and accessed at any time.
- These states are called as 'commits' in git's terminology



The basic use of git

Record snapshots of how a project develop

- Only the committed stages are recorded, the rest of the history is discarded
- The git repository is recorded as changes from one commit to the next

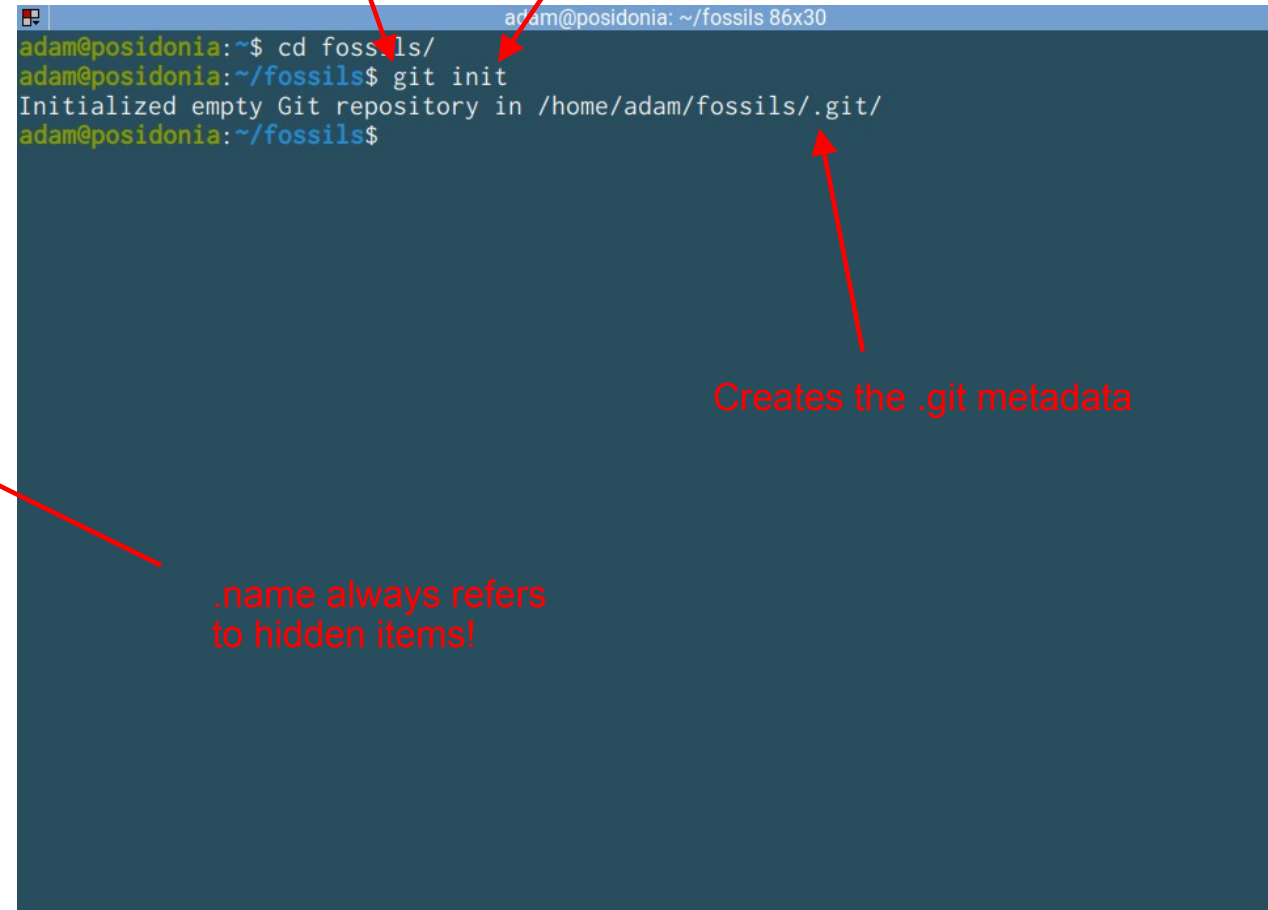


git_init

Create a new git repository in current directory.

- A git repository is a directory with git metadata in it.
- The git metadata are in the `.git` directory

Name of application Command for the application



```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~$ cd fossils/
adam@posidonia:~/fossils$ git init
Initialized empty Git repository in /home/adam/fossils/.git/
adam@posidonia:~/fossils$
```

Annotations:

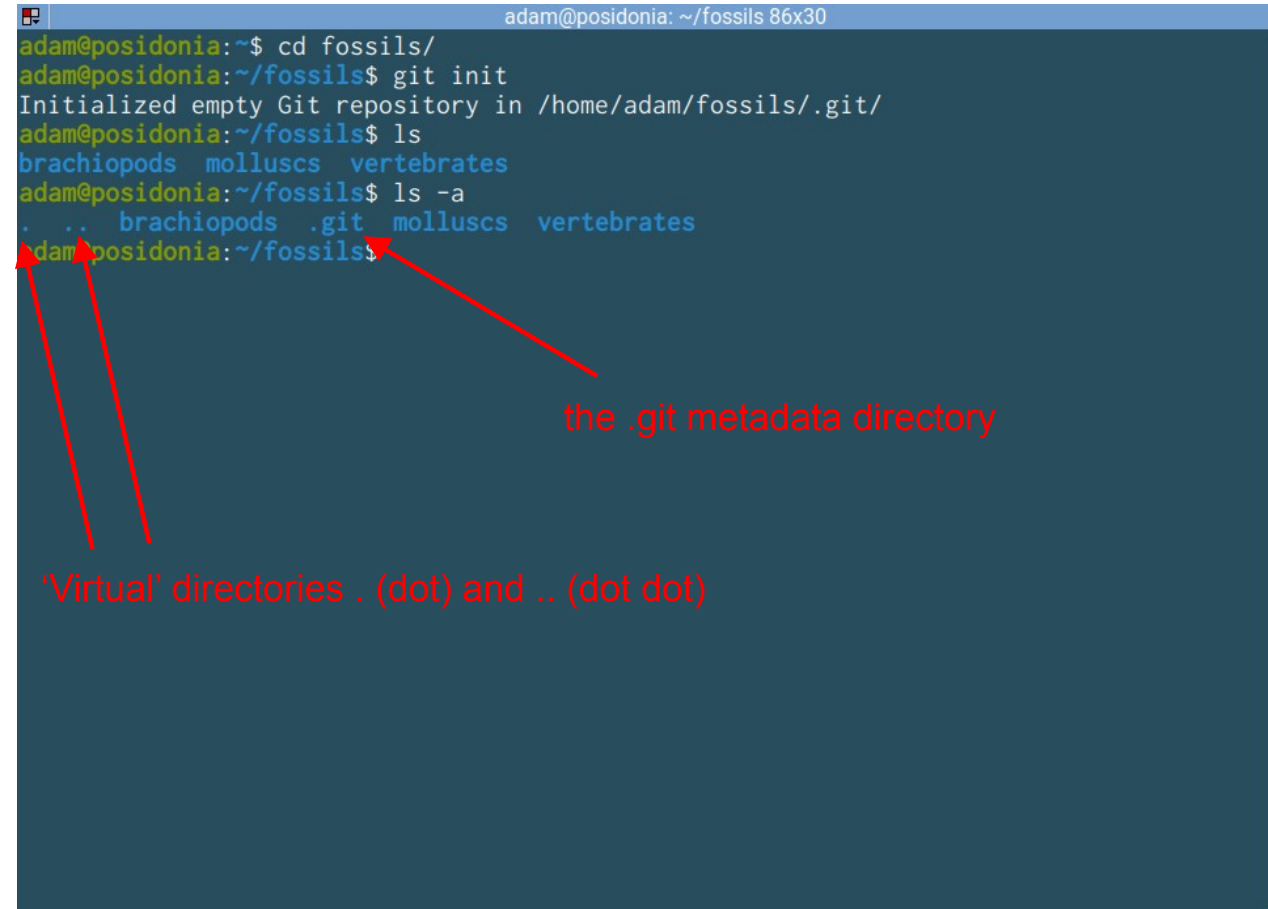
- Red arrow from "Name of application" points to `fossils` in `cd fossils/`.
- Red arrow from "Command for the application" points to `git init`.
- Red arrow from "Creates the .git metadata" points to `/home/adam/fossils/.git/` in the output.
- Red arrow from "The git metadata are in the `.git` directory" points to `.git` in the output.
- Red text at the bottom: `.name always refers to hidden items!`

ls -a

List all files and directories in directory, including hidden items!

- The double dot (..) represents a way to refer to the previous directory, as we have seen earlier
- The single dot (.) represents a way to refer to the current directory.
- **Note:** `cd brachiopods` and `cd ./brachiopods` are the same!

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~$ cd fossils/
adam@posidonia:~/fossils$ git init
Initialized empty Git repository in /home/adam/fossils/.git/
adam@posidonia:~/fossils$ ls
brachiopods  molluscs  vertebrates
adam@posidonia:~/fossils$ ls -a
.  ..  brachiopods  .git  molluscs  vertebrates
adam@posidonia:~/fossils$
```



git_status

Show the status of the current repository

- A series of commits is called a 'branch'. Simple repos use only one. There is always a current one
- Git has detected that there are things in the repo that are not registered.
- **Git can only detect files. Empty directories are not recorded!**

The image shows a terminal window with the command `git status` executed. The output is as follows:

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    brachiopods/

nothing added to commit but untracked files present (use "git add" to track)
adam@posidonia:~/fossils$
```

Annotations with red arrows point to specific parts of the terminal output:

- Name of the current "branch"**: Points to `On branch master`.
- Name of application**: Points to `git` in the command `git status`.
- Command for the application**: Points to `status` in the command `git status`.
- We have no commits yet**: Points to `No commits yet`.
- NOTE: In many cases, git literally tells you what to do.**: Points to the instruction `(use "git add" to track)`.

Staging

The preparation of a commit

- Commits are permanent, or are difficult to remove once done, so we have tools to make sure that they are ok
- Changes first have to be staged, before committing. This allows us to include only specific changes in the commit, and to make sure that we are doing things ok.

Staging and commit (Airport)

Initial boarding pass control vs. boarding

- If you go through security you are staged to fly. You are expected to be on the plane, but you can still leave.
- If you board the plane and the cabin doors are closed, you are committed to a flight.

Getting staged



In the staged area, waiting to be committed



The commit

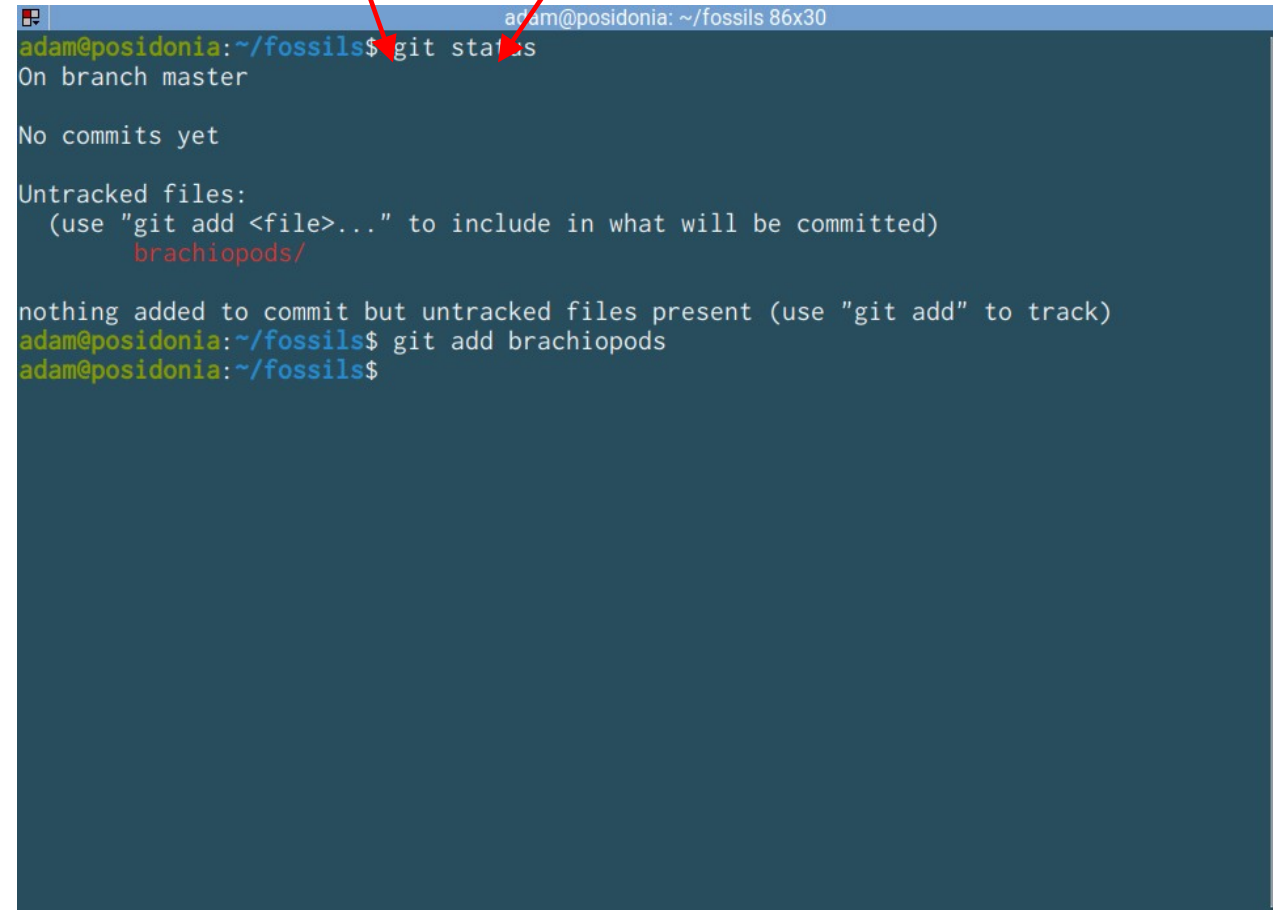


`git_add_<path>`

Stage the target file or directory.

- Frequently this is an entire directory, including . (dot)
- If successful does not return anything, has to be checked with git status

Name of application Command for the application



```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    brachiopods/

nothing added to commit but untracked files present (use "git add" to track)
adam@posidonia:~/fossils$ git add brachiopods
adam@posidonia:~/fossils$
```

The terminal screenshot shows the execution of `git status` and `git add brachiopods`. Two red arrows originate from labels above the terminal: one points to the `git` command in the first line, and the other points to the `brachiopods` argument in the second line.

git_status (again)

Show status of repo

- There is just one file here which git finds.
- The file is now staged to be committed.

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    brachiopods/

nothing added to commit but untracked files present (use "git add" to track)
adam@posidonia:~/fossils$ git add brachiopods
adam@posidonia:~/fossils$ echo $?
0
adam@posidonia:~/fossils$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   brachiopods/genera.txt

adam@posidonia:~/fossils$
```

Things really went error free
(not necessary to check)

If you have changed your
mind, do what git tells you!

git commit -m <message>

First use not permitted without credentials!

- You need to provide a user name and an email address with the git config command

```
File Machine View Input Devices Help
MINGW64/c/Users/Adam/fossils

Adam@Teaching MINGW64 ~/fossils (master)
$ git commit -m "First file added"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Adam@Teaching.(none)')

Adam@Teaching MINGW64 ~/fossils (master)
$ |
```


git_config_--global_<what>_<value>

Configuring git

- user.name and user.email
- --global sets this for all your local git repositories
- Now you are ready to commit

```
File Machine View Input Devices Help
MINGW64/c/Users/Adam/fossils

Adam@Teaching MINGW64 ~/fossils (master)
$ git commit -m "First file added"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Adam@Teaching.(none)')

Adam@Teaching MINGW64 ~/fossils (master)
$ git config --global user.email "adam.kocsis@outlook.com"

Adam@Teaching MINGW64 ~/fossils (master)
$ git config --global user.name "adamkocsis"

Adam@Teaching MINGW64 ~/fossils (master)
$
```


`git_commit_-m_<message>`

Now create a new commit

- Provide a message in **quotes!**
This is the human readable description of what changed.
- Every commit gets a unique 'hash', a random set of characters that are used to identify unambiguously identify the commit

The screenshot shows a terminal window with the following text:

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ git commit -m "First file added."
[master (root-commit) 6c61587] First file added.
1 file changed, 4 insertions(+)
create mode 100644 brachiopods/genera.txt
adam@posidonia:~/fossils$
```

Red arrows point to specific parts of the output with the following labels:

- The message you provided**: Points to the commit message "First file added."
- The beginning of the hash of the commit.**: Points to the hash "6c61587".
- branch**: Points to the branch name "master".
- Four new lines are added**: Points to the line "1 file changed, 4 insertions(+)"
- The affected files**: Points to the file "brachiopods/genera.txt".

git_status (yet again)

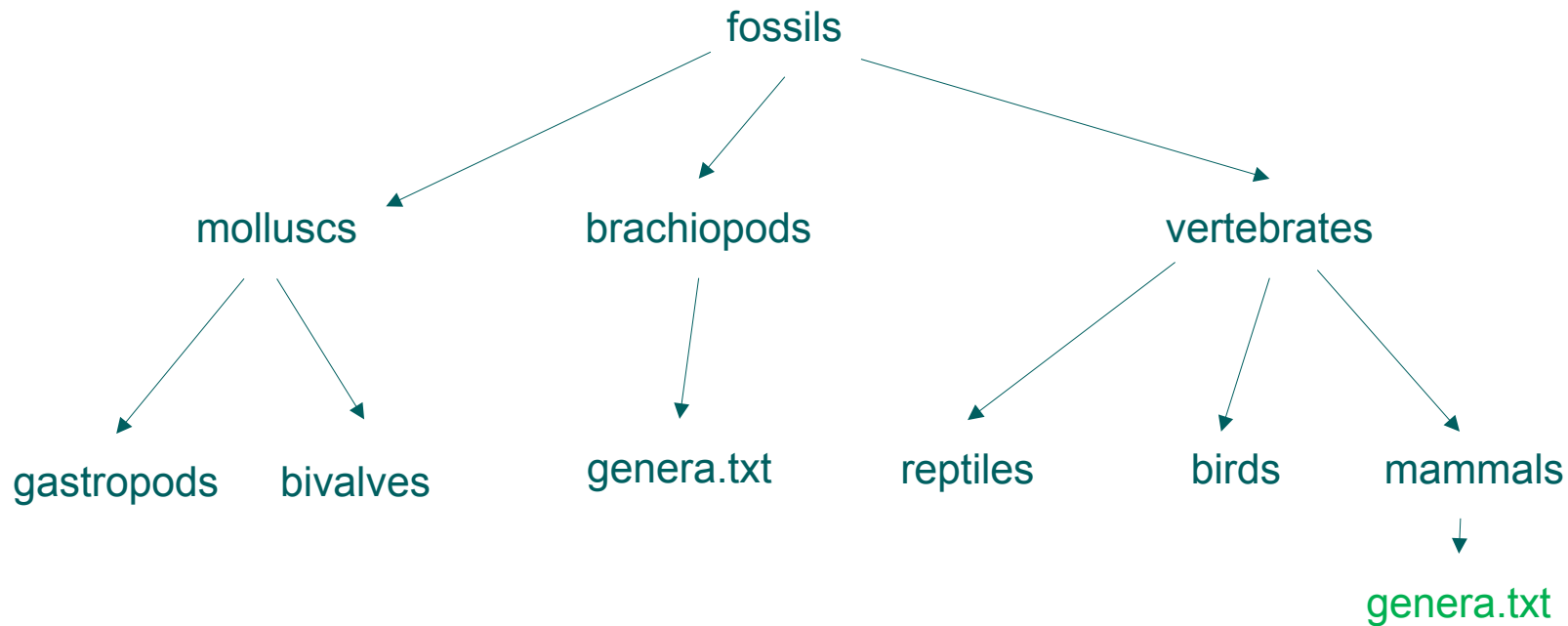
Nothing to be done.

- Create two new files

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ git commit -m "First file added."
[master (root-commit) 6c6158e] First file added.
1 file changed, 4 insertions(+)
create mode 100644 brachiopods/genera.txt
adam@posidonia:~/fossils$ git status
On branch master
nothing to commit, working tree clean
adam@posidonia:~/fossils$
```

Demo!

1. Create a new file `genera.txt` in the `mammals` directory, and put the names of 3 mammalian genera in it!
2. Stage and commit the changes!



My solution

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ echo -e "Mustela\nHomo\nPanthera" > ./vertebrates/mammals/genera.txt
adam@posidonia:~/fossils$ cat vertebrates/mammals/genera.txt
Mustela
Homo
Panthera
adam@posidonia:~/fossils$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    vertebrates/

nothing added to commit but untracked files present (use "git add" to track)
adam@posidonia:~/fossils$ git add .
adam@posidonia:~/fossils$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   vertebrates/mammals/genera.txt

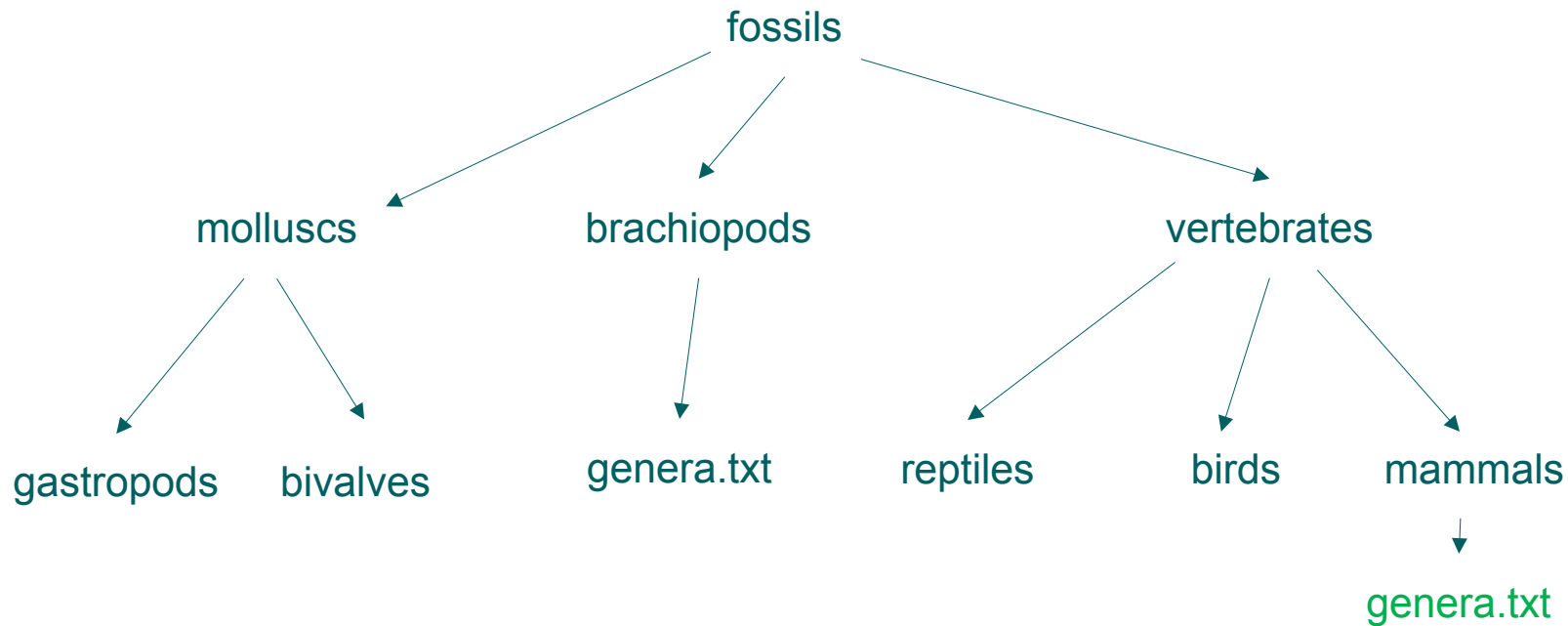
adam@posidonia:~/fossils$ git commit -m "added vertebrate genera"
[master 510177f] added vertebrate genera
1 file changed, 3 insertions(+)
create mode 100644 vertebrates/mammals/genera.txt
adam@posidonia:~/fossils$
```

Contents of the new file

Add everything you find in current directory.

Demo!

1. Create a new file `genera.txt` in the `birds` directory, and put the names of 2 bird genera in it!
2. Add another genus to the mammals.
3. Try to commit only the birds!



My solution

1. Make the changes.

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ echo -e "Pica\nTurdus" > "vertebrates/birds/genera.txt"
adam@posidonia:~/fossils$ cat vertebrates/birds/genera.txt
Pica
Turdus
adam@posidonia:~/fossils$ echo "Talpa" >> vertebrates/mammals/genera.txt
adam@posidonia:~/fossils$ cat vertebrates/mammals/genera.txt
Mustela
Homo
Panthera
Talpa
adam@posidonia:~/fossils$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vertebrates/mammals/genera.txt
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        vertebrates/birds/
no changes added to commit (use "git add" and/or "git commit -a")
adam@posidonia:~/fossils$
```

← Add birds

← Added another mamal

← Change in already committed file

← New entries to be added

My solution

2. Stage only the birds.

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ cat vertebrates/mammals/genera.txt
Mustela
Homo
Panthera
Talpa
adam@posidonia:~/fossils$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vertebrates/mammals/genera.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        vertebrates/birds/

no changes added to commit (use "git add" and/or "git commit -a")
adam@posidonia:~/fossils$ git add vertebrates/birds
adam@posidonia:~/fossils$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   vertebrates/birds/genera.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vertebrates/mammals/genera.txt

adam@posidonia:~/fossils$
```

Staged.

Not staged!

My solution

3. Stage only the birds.

```
adam@posidonia: ~/fossils 86x30
Talpa
adam@posidonia:~/fossils$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vertebrates/mammals/genera.txt

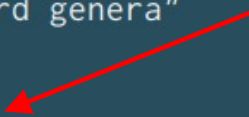
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        vertebrates/birds/

no changes added to commit (use "git add" and/or "git commit -a")
adam@posidonia:~/fossils$ git add vertebrates/birds
adam@posidonia:~/fossils$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   vertebrates/birds/genera.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vertebrates/mammals/genera.txt

adam@posidonia:~/fossils$ git commit -m "added bird genera"
[master b53f2f9] added bird genera
1 file changed, 2 insertions(+)
create mode 100644 vertebrates/birds/genera.txt
adam@posidonia:~/fossils$
```

Nothing happened to mammals!



git_restore_<path>

Discarding changes from previous commit

- We can commit the new mammal or discard it.
- You can correct unintended changes with this.
- What about even older changes?

Again, git literally tells you your options

```
adam@posidonia: ~/fossils 86x30
adam@posidonia:~/fossils$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   vertebrates/mammals/genera.txt

no changes added to commit (use "git add" and/or "git commit -a")
adam@posidonia:~/fossils$ git restore vertebrates/mammals/genera.txt
adam@posidonia:~/fossils$ cat vertebrates/mammals/genera.txt
Mustela
Homo
Panthera
adam@posidonia:~/fossils$
```

The file is restored to the state before the changes, what is in the commit.

git_log

Accessing repository history

- Commits listed out in sequence.
- Every commit has a hexadecimal hash id
- Lists out metadata of the commit
- The 'HEAD' indicates where we are in the history

```
commit 24cd260be44f38ac80994594cf4e048b294be660 (HEAD → master)
Author: Ádám T. Kocsis <adam.kocsis@fau.de>
Date:   Mon Aug 18 05:41:53 2025 +0200

    added bird genera

commit 25e3ad4c4446aaf98151ed84db874494bbf04aef
Author: Ádám T. Kocsis <adam.kocsis@fau.de>
Date:   Mon Aug 18 05:40:20 2025 +0200

    Added vertebrates

commit 00bf66d0064396091a8fc1dc36df7b355c5398c9
Author: Ádám T. Kocsis <adam.kocsis@fau.de>
Date:   Mon Aug 18 05:38:45 2025 +0200

    Initial
(END)
```

git_checkout_<hash>

Exploring history

- No need to write the entire hash
- Moves back in time so we can explore the past states
- `git_checkout_<branch name>` moves the head back to the most recent commit

birds/genera.txt is not there! →

Move head back →

birds/genera.txt reappears! →

```
adam@TheStuff ~/work/fossils master git log
adam@TheStuff ~/work/fossils master git checkout 25e3
Note: switching to '25e3'.
```

You are in 'detached HEAD' state. You can look around, make experimental changes and commit them, and you can discard any commits you make in this state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may do so (now or later) by using `-c` with the switch command. Example:

```
git switch -c <new-branch-name>
```

Or undo this operation with:

```
git switch -
```

Turn off this advice by setting config variable `advice.detachedHead` to false

HEAD is now at 25e3ad4 Added vertebrates

```
adam@TheStuff ~/work/fossils - 25e3ad4 ls vertebrates
mammals reptiles
```

```
adam@TheStuff ~/work/fossils - 25e3ad4 git checkout master
Previous HEAD position was 25e3ad4 Added vertebrates
Switched to branch 'master'
```

```
adam@TheStuff ~/work/fossils master ls vertebrates
birds mammals reptiles
adam@TheStuff ~/work/fossils master |
```

GitHub

and GitHub

GitHub

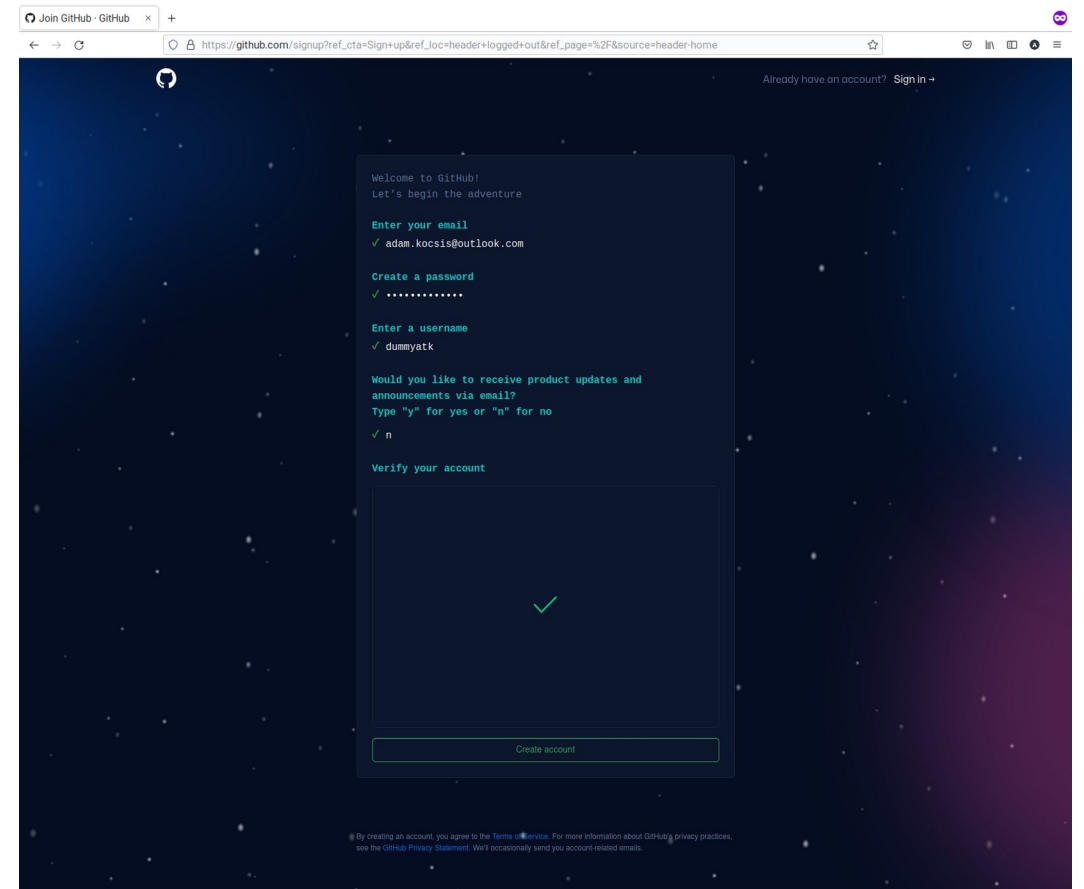
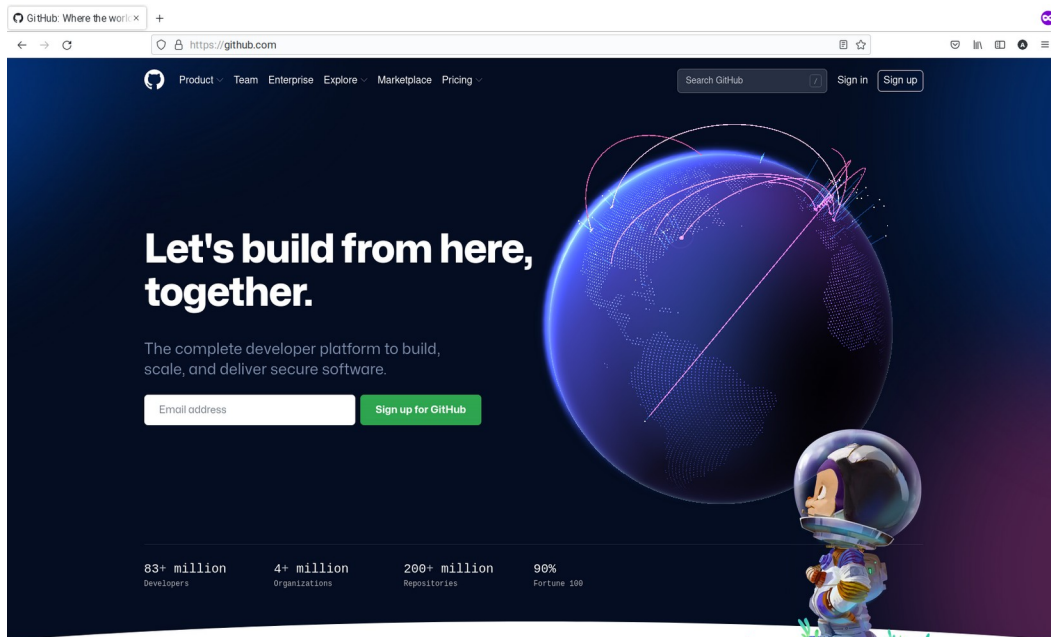
Where the world builds software (2008-)

- Open source software development platform, places to store and share git repositories
- Currently owned by Microsoft
- Applications, packages, plugins, webpages and many more!
- Free and private repositories.
- Tools for CI/CD (continuous integration/deployment)



GitHub

Sign up if you haven't yet!



GitHub - Dashboard

The image shows the GitHub dashboard interface. At the top, a browser window displays the URL `https://github.com`. The dashboard header includes a search bar, navigation links for Pull requests, Issues, Marketplace, and Explore, and a user profile section with a notification bell and a settings gear icon. Two red arrows point to these icons, both labeled "Notifications".

Notifications

Notifications

Settings

Create your first project
Ready to start building? Create a repository for a new idea or bring over an existing repository to keep contributing to it.
[Create repository](#) [Import repository](#)

Recent activity
When you take actions across GitHub, we'll provide links to that activity here.

The home for all developers — including you.
Welcome to your personal dashboard, where you can find an introduction to how GitHub works, tools to help you build software, and help merging your first lines of code.

Start writing code

- Start a new repository**
Collaborate on code with others and track your work in a repository.
[Create a new repository](#)
- Create your profile README**
Create a file in a repository that tells the GitHub community who you are.
[Create a README](#)
- Contribute to an existing repository**
[Find repos that need your help >](#)

Use tools of the trade

- Write code in your web browser**
Use the [github.dev](#) web-based editor from your repository or pull request to create and commit changes.
- Install a powerful code editor**
Visual Studio Code is a multi-platform code editor optimized for building and debugging software.
- Set up your local dev environment**
After you [set up Git](#), simplify your dev workflow with [GitHub Desktop](#), or [bring GitHub to the command line](#).

GitHub Copilot
Get suggestions for lines of code and entire functions in real-time.
[Learn more about Copilot](#)

PRIVACY STATEMENT UPDATES
Adding web cookies for enterprise users
In order to better reach and improve the web experience for enterprise users, we are adding non-essential web cookies to certain subdomains that specifically market our products to businesses. This change is only on subdomains that reach enterprise customers, and all other GitHub subdomains will continue to operate as-is.
[Learn more](#)

Latest changes

GitHub - Dashboard

Access your repos

Settings

The screenshot shows the GitHub dashboard interface. At the top, the browser address bar displays 'https://github.com'. The navigation bar includes a search bar, links for 'Pull requests', 'Issues', 'Marketplace', and 'Explore', and a user profile icon. A red arrow points from the text 'Access your repos' to the 'Your repositories' option in the user menu. Another red arrow points from the text 'Settings' to the 'Settings' option in the same menu. The main content area is divided into several sections: 'Create your first project' with buttons for 'Create repository' and 'Import repository'; 'Recent activity'; 'The home for all developers — including you.' with a welcome message and links to 'Start writing code', 'Start a new repository', 'Create your profile README', 'Contribute to an existing repository', 'Write code in your web browser', 'Install a powerful code editor', and 'Set up your local dev environment'; and a 'Latest changes' section at the bottom. A sidebar on the right features 'GitHub Copilot' and a 'Privacy Statement Updates' section.

GitHub

Search or jump to...

Pull requests Issues Marketplace Explore

Create your first project

Ready to start building? Create a repository for a new idea or bring over an existing repository to keep contributing to it.

Create repository Import repository

Recent activity

When you take actions across GitHub, we'll provide links to that activity here.

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Find repos that need your help >

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Learn more

Signed in as dummyatk

Set status

Your profile

Your repositories

Your codespaces

Your projects

Your stars

Your gists

Upgrade

Feature preview

Help

Settings

Sign out

Latest changes

GitHub – Creating a new repo

Make a new repo

The screenshot shows the GitHub interface for a user named 'dummyatk'. At the top, there's a navigation bar with 'Your Repositories' and a search bar. Below this, the user's profile is displayed, including a circular profile picture and the username 'dummyatk'. The 'Repositories' tab is selected, showing a search bar and filters for 'Type', 'Language', and 'Sort'. A green 'New' button is visible on the right. A red arrow points from the text 'Make a new repo' to this button. The main content area states 'dummyatk doesn't have any public repositories yet.' The footer contains the GitHub logo, copyright information, and various links like 'Terms', 'Privacy', 'Security', etc.

Your Repositories x +

https://github.com/dummyatk?tab=repositories

Search or jump to... Pull requests Issues Marketplace Explore

Overview Repositories Projects Packages Stars

Find a repository... Type Language Sort New

dummyatk

Edit profile

Joined 7 minutes ago

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GitHub – Creating a new repo



Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Owner *

 dummyatk

Repository name *

fossils

Great repository names are short and memorable. Need inspiration? How about [shiny-meme?](#)

Description (optional)

Just an exercise.

☒ Public

Anyone on the internet can see this repository. You choose who can commit.

☐ Private

You choose who can see and commit to this repository.

Initialize this repository with:

Skip this step if you're importing an existing repository.

☐ Add a README file

This is where you can write a long description for your project. [Learn more.](#)

Add .gitignore

Choose which files not to track from a list of templates. [Learn more.](#)

.gitignore template: None

Choose a license

A license tells others what they can and can't do with your code. [Learn more.](#)

License: None

 You are creating a public repository in your personal account.

Create repository

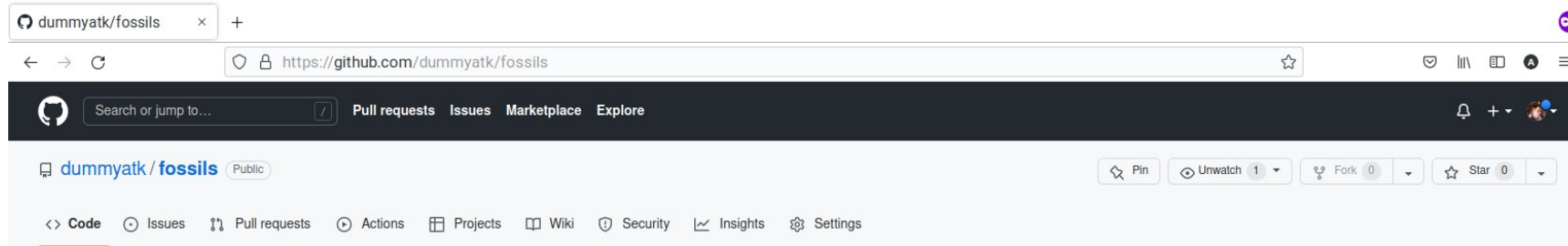
Usually the same as the local directory.

Do not change these. You will copy files over from your local repo.

Things for others!

Create!

GitHub – The fresh empty repo



This is what you want

Quick setup — if you've done this kind of thing before

or ☐ HTTPS ☐ SSH

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# fossils" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/dummyatk/fossils.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/dummyatk/fossils.git
git branch -M main
git push -u origin main
```

...or import code from another repository

You can initialize this repository with code from a Subversion, Mercurial, or TFS project.

[Import code](#)

We have already done this mostly

Mac Users

- Can only use ssh to authenticate!

GitHub – Add new remote

The name of the new remote. You can refer to it from now on using this name!

The URL of the remote. This is used to identify the remote on the web.

...or push an existing repository from the command line

```
git remote add origin https://github.com/dummyatk/fossils.git
git branch -M main
git push -u origin main
```

Application name

Command: you want to make changes of how your local repository is connected to remotes

You are registering a new remote

GitHub – Rename current branch to main

Application name Command: you want to do things with branches Move all contents of current branch to The name of the branch (new)

...or push an existing repository from the command line

```
git remote add origin https://github.com/dummyatk/fossils.git
git branch -M main
git push -u origin main
```

For political reasons, GitHub does not allow the use of the name master, hence this extra step.

GitHub – Pushing contents of branch to remote

Application name

Command: you want copy
contents from local to
remote

Set the default remote and branch

Remote to copy material to

Which branch to push?

...or push an existing repository from the command line

```
git remote add origin https://github.com/dummyatk/fossils.git
git branch -M main
git push -u origin main
```

GitHub will ask for your credentials

GitHub – Executing this and signing in on windows

File Machine View Input Devices Help

MINGW64: c:/Users/Adam/fossils

```
Adam@Teaching MINGW64 ~/fossils (master)
$ git remote add origin https://github.com/dummyatk/fossils.git

Adam@Teaching MINGW64 ~/fossils (master)
$ git branch -M main

Adam@Teaching MINGW64 ~/fossils (main)
$ git push -u origin main
```

Note branch name change

Most interactive sign
in option available on
Windows

Connect to GitHub

GitHub
Sign in

Browser/Device Token

Sign in with your browser

Sign in with a code

Don't have an account? [Sign Up](#)

GitHub – Executing this and signing in on windows



Sign in to **GitHub**
to continue to **Git Credential Manager**

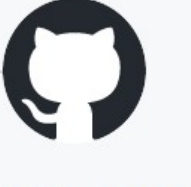
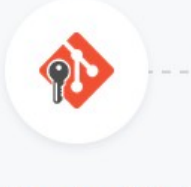
Username or email address

Password [Forgot password?](#)

Sign in

New to GitHub? [Create an account](#) .

This is what you want



Authorize Git Credential Manager

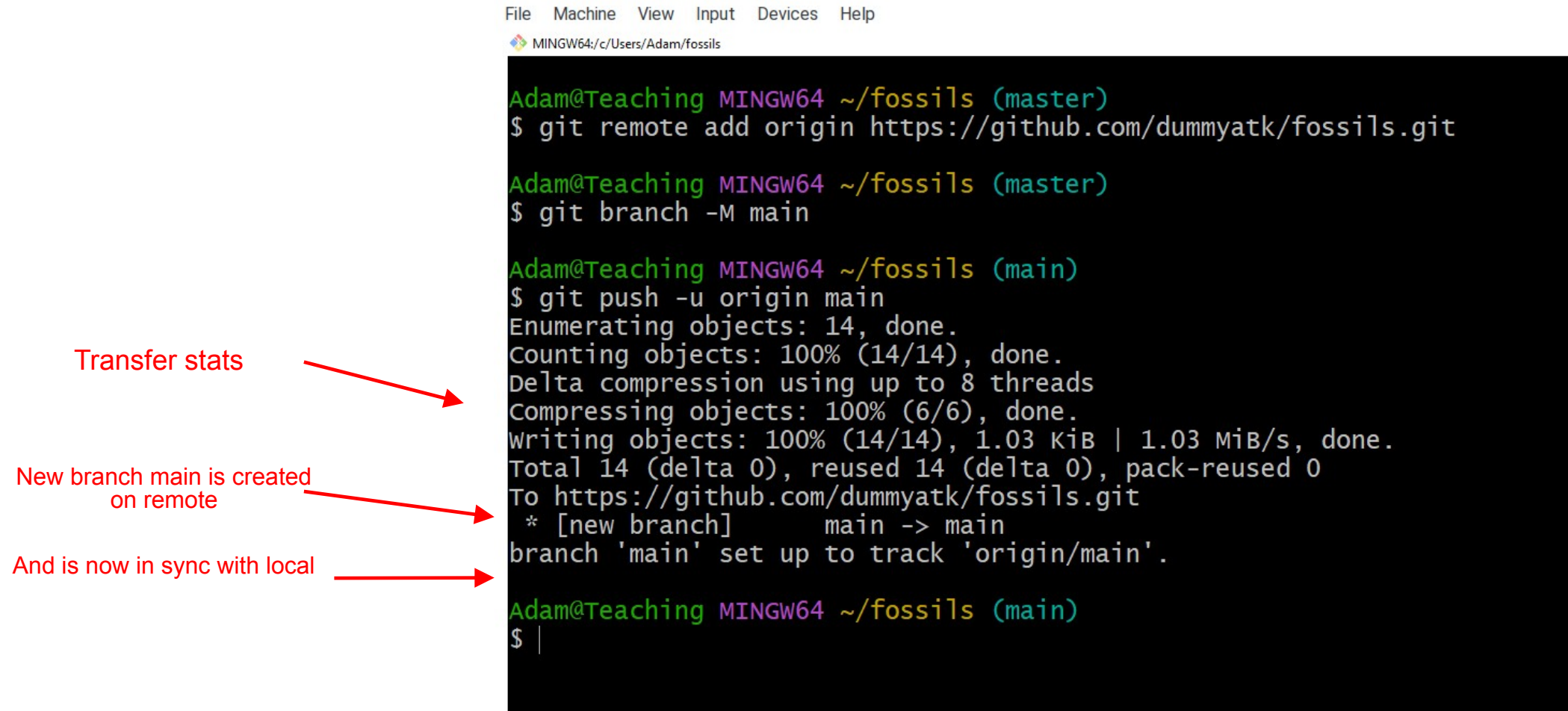
 **Git Credential Manager** by [GitCredentialManager](#)
wants to access your dummyatk account

-  **Gists**
Read and write access
-  **Repositories**
Public and private
-  **Workflow**
Update GitHub Action Workflow files.

Cancel **Authorize GitCredentialManager**

Authorizing will redirect to
<http://localhost:50855>

GitHub – Successful push



```
File Machine View Input Devices Help
MINGW64/c/Users/Adam/fossils

Adam@Teaching MINGW64 ~/fossils (master)
$ git remote add origin https://github.com/dummyatk/fossils.git

Adam@Teaching MINGW64 ~/fossils (master)
$ git branch -M main

Adam@Teaching MINGW64 ~/fossils (main)
$ git push -u origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (14/14), 1.03 KiB | 1.03 MiB/s, done.
Total 14 (delta 0), reused 14 (delta 0), pack-reused 0
To https://github.com/dummyatk/fossils.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Adam@Teaching MINGW64 ~/fossils (main)
$ |
```

Transfer stats

New branch main is created on remote

And is now in sync with local

GitHub – Successful push

The screenshot shows the GitHub interface for the repository 'dummyatk/fossils'. The top navigation bar includes a search bar, links to Pull requests, Issues, Marketplace, and Explore, and a user profile icon. The repository header shows the name 'dummyatk/fossils' as Public, with buttons for Pin, Unwatch (1), Fork (0), and Star (0). Below the header, a navigation bar contains links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. The main content area displays the 'main' branch with 1 branch and 0 tags. A commit by 'adamkocsis' is shown, titled 'added bird genera', with a file list: 'brachiopods' (First file added, 2 hours ago) and 'vertebrates' (added bird genera, 1 hour ago). A light blue banner prompts the user to 'Help people interested in this repository understand your project by adding a README.' with a green 'Add a README' button. A red arrow points from the text 'A Readme is quite useful' to this button. The right sidebar contains sections for 'About' (Just an exercise, 0 stars, 1 watching, 0 forks), 'Releases' (No releases published, Create a new release), and 'Packages' (No packages published, Publish your first package). The footer includes the GitHub logo, copyright notice '© 2022 GitHub, Inc.', and various links like Terms, Privacy, Security, Status, Docs, Contact GitHub, Pricing, API, Training, Blog, and About.

Search or jump to... Pull requests Issues Marketplace Explore

dummyatk / fossils Public

Pin Unwatch 1 Fork 0 Star 0

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

main 1 branch 0 tags

Go to file Add file Code

adamkocsis added bird genera b53f2f9 1 hour ago 3 commits

brachiopods	First file added.	2 hours ago
vertebrates	added bird genera	1 hour ago

Help people interested in this repository understand your project by adding a README. Add a README

A Readme is quite useful

About Just an exercise. 0 stars 1 watching 0 forks

Releases No releases published Create a new release

Packages No packages published Publish your first package

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Writing a Readme

- Default format is **markdown (later)**
- You can work on files using GitHub's interface
- Save the defaults, by clicking on the green button
- Note that you are technically creating a new commit!

The screenshot shows the GitHub web interface for creating a new file. The repository is 'dummyatk / fossils'. The file being created is 'README.md' in the 'main' branch. The editor shows the following content:

```
1 # fossils
2 Just an exercise.
3
```

Red arrows and text annotations highlight key features:

- An arrow points to the 'Preview' tab, with the text: "File name: .md is for markdown."
- An arrow points to the default content in the editor, with the text: "Some default content, we will visit this again."
- An arrow points to the 'Commit new file' button, with the text: "Default commit message."

The 'Commit new file' section shows the commit message 'Create README.md' and an optional extended description field. The commit is set to be made directly to the 'main' branch.

GitHub – Changing the remote

The very last commit's hash

The screenshot displays the GitHub interface for the repository 'dummyatk/fossils'. The repository is public and has 1 branch and 0 tags. The file list shows 'brachiopods', 'vertebrates', and 'README.md'. The 'README.md' file is highlighted, and its content is displayed below. The content is a beautifully rendered markdown document with the title 'fossils' and the text 'Just an exercise.'.

Annotations on the image include:

- A red arrow pointing to the 'Add file' button.
- A red arrow pointing to the 'README.md' file in the file list.
- A red arrow pointing to the 'README.md' file content, with the text 'Beautifully rendered markdown document'.
- A red arrow pointing to the 'README.md' file content, with the text 'The message of last commit that modified the file'.

The message of the last commit that modified the file

Readme file now added!

Beautifully rendered markdown document

git_pull_<remote>_<branch>

Pull changes from remote

- Just because you changed something on the remote server does not make things magically appear locally
- You have to pull the contents of the remote to have the new file that you just created!

File Machine View Input Devices Help

MINGW64:/c/Users/Adam/fossils

```
Adam@Teaching MINGW64 ~/fossils (main)
$ git pull origin main
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (3/3), 715 bytes | 55.00 KiB/s, done.
From https://github.com/dummyatk/fossils
 * branch                main          -> FETCH_HEAD
    b53f2f9..a4a30cd      main          -> origin/main
Updating b53f2f9..a4a30cd
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 README.md

Adam@Teaching MINGW64 ~/fossils (main)
$ cat README.md
# fossils
Just an exercise.

Adam@Teaching MINGW64 ~/fossils (main)
$ |
```